



**National University of Computer and Emerging Sciences, Lahore**



# **JobLens - NLP Enabled Deep Learning Approach for Accurate Resume Ranking**

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## Abstract

The flood of employment applications and the variety of CV formats constitute a difficult problem in the dynamic world of human resources (HR) and recruitment. Finding individuals whose qualifications match particular job needs quickly and accurately is a difficult assignment for HR professionals. Conventional methods for screening resumes are frequently ineffective, requiring significant time and resource expenditures. As a result, a comprehensive solution that can streamline the hiring process and guarantee the selection of the best candidates becomes urgently necessary.

Our study intends to construct an intelligent resume parsing and ranking system in answer to this difficulty by utilizing Natural Language Processing (NLP) and Machine Learning (ML) approaches. By giving HR professionals a strong tool to effectively match resumes to job descriptions, this system will revolutionize the way they assess and choose candidates. Our system will be able to better grasp contextual information, increase accuracy, and adjust to changing resume rating criteria by utilizing NLP and ML algorithms.

A large dataset of around 1000+ resumes has been made by collecting data from various sources. Different pre-processing techniques have been applied to standardize and normalize the dataset, moreover it has been annotated to train the NLP module. A deep learning model was trained on this dataset to identify key features. The system was tested vigorously, evaluating its accuracy and efficiency and around 96% accuracy was achieved. With this the ranking module took input from the NLP module and of job description from the user. Finally the similarity index was calculated between the description and the resumes. The highest rank was given to the resumes with the best similarity index. Metric used for this was Normalized Discounted Cumulative Gain (NDCG) metric to evaluate the quality of the ranking algorithm. This metric gives a range from 0 to 1. The average NDCG value achieved for the ranking module was 0.85. To conclude, the research's objectives, and scope help identify its potential to bring positive change to the recruitment process.

## Executive Summary

In today's developing information technology job market, the process of finding or identifying the right person for a job position has become very tiring and time-consuming. The methods being used by recruiters nowadays are very manual meaning they lack efficiency and accuracy and are often biased leading to wrong decisions while hiring a candidate.

The project vision is to develop a product that uses deep learning algorithms and Natural Language Processing (NLP) techniques to extract useful information from the resumes and using the extracted information to rank resumes based on criteria defined by recruiters, which we will be calling "JobLens". This approach will assist the hiring process, reduce bias, and help software houses to make better decisions. A large dataset of resumes has been made by collecting data from various sources and different pre-processing techniques have been applied to standardize and normalize the dataset, moreover we have ourselves annotated a set of around 1400 resumes to train our NLP module. Details are further discussed in dataset section.

Under the section of proposed methodology and approach, in-depth description of the modules has been given. A deep learning model will be trained on this dataset to identify key features. The system will undergo testing, evaluating its accuracy and efficiency. With this our ranking module will be taking input from our previous module and input of job description from the user. Finally the similarity index will be calculated between the description and the resumes. The highest rank will be given to the resume with the best similarity index.

Moreover we implemented the NLP and resume ranking modules that we had built and ran a few experiments on them and obtained promising results. In short we were able to obtain 96 percent accuracy for our NLP module meaning that the resumes parsed with our module will extract the useful information from the resumes very accurately. With this our ranking module also gave promising results in which we were able to correctly identify the most similar resume to the job description. Our project aligns with the sustainable development goal of decent work and economic growth. By improving the resume screening process, "JobLens" will help organizations to make more efficient hiring decisions, resulting in quicker job placements that match candidates' qualifications. Its efficiency supports economic growth by ensuring individuals are employed in positions that align with their skills and experience.

To conclude, the project's objectives, scope, and alignment with sustainable development goal help identify its potential to bring positive change to the recruitment process.

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## Chapter 1 Introduction

The Human Resource department is a crucial part of any organization. HR's responsibility is to make decisions considering someone "Best" for a job. An organization's success relies on the performance of their employees hence it is very important to hire the right person. Due to the easy availability of information and resources on the internet, many people have become skilled enough to look for a job. Imagine being an HR in a multinational firm and being asked to review thousands of resumes for a single job vacancy. It would be tiring and difficult to skim through all the resumes and decide to hire a person. There is a high probability that you make an unfair or wrong decision and end up hiring someone who is not capable enough to do your job. Imagine you could have hired a person who was far better than the person you hired but the limit of being a human led you to this point. Why choose the average when you can choose the best?

To overcome all these obstacles and end up with self-satisfaction regarding your decision to hire someone, we decided to make "JobLens". It is specifically designed to facilitate the Human Resources (HR) department of software/IT firms in terms of performing initial screening/filtering of resumes to assist their hiring process. JobLens is a web-based application that will utilize deep learning algorithms to automate the resume screening process and make it hassle-free. It will use NLP (Natural Language Processing) to extract relevant information from a resume to train a deep learning model. Web-based applications will allow the user to upload their resumes and input the required parameters. The trained deep-learning model will rank the resumes based on the input parameters and present us with the best-fit resume(s). This way a fair decision could be made with minimum human effort and time.

### 1.1 Purpose of this Document

The purpose of this document is to present our approach towards formulating a possible solution to perform an initial screening of the CV/resume by parsing relevant data like skills, education, and experience, and performing ranking based on a specific benchmark. It is a web-based application that has the potential to facilitate the HR(Human Resources) department of IT/Software organizations to perform the initial screening/filtering of CV's for their hiring process for a particular job post. The main aim of this project is to develop a user-friendly MVP(Minimum Viable Product) that allows users to upload multiple resumes of different formats. Our project is NLP(Natural Learning Processing) domain-oriented since it uses models that are able to perform the magic of resume parsing and ranking to deliver the best possible results it can. The research question that we aim to answer is: " Can we develop a web-based application with improved accuracy for resume parsing and then use the parsed information to perform resume ranking in order to facilitate the process of hiring?". This report presents the methodology adopted to

reach a potential solution. It mainly highlights the data sets, models, training and testing, design, and implementation details of our project.

## 1.2 Intended Audience

The intended audience for this project is HR(Human Resource) professionals of software houses or IT-related firms because our project is a tool for them to boost their performance and quality of work. If they want a helping hand to assist them in the hiring process they can use our project to perform the initial screening/filtering of resumes and rank them based on their custom requirements like experience, skills, and other relevant criteria. This will help them save their time, and effort and choose suitable candidates to some extent.

## 1.3 Definitions, Acronyms, and Abbreviations

- **NLP:** Natural Language Processing
- **HR:** Human Resources
- **MVP:** Minimum Viable Product
- **CV:** Curriculum Vitae
- **IT:** Information Technology
- **SDG:** Sustainable Development Goal
- **FYP:** Final Year Project

## 1.4 Conclusion

To conclude, the first chapter of our document gives you a brief overview of our project "JobLens" which aims to assist HR's of IT and software-related firms in the hiring process. It is a web and NLP-based application that allows HR to upload resumes in order to get them parsed and ranked to perform the initial filtering in their hiring process. Moreover, it describes the purpose of this document and the target audience for which this piece of information will be a valuable insight. The second chapter of this document, Project Vision defines the problem domain overview and further elaborates it by relating it to the real-world problem of the tiring and hectic recruitment process. Moreover, this section also points down the goals and objectives of our project and clearly mentions the project scope. The decent work and economic growth sustainable development goal is targeted by our project, also covered in this section. The third chapter of our document, Literature Review refers to some notable work that is similar

in nature to our domain of project. It includes 21 quality published documents which collectively highlights methodologies, data sets and models used to achieve their respective desired applications. We have also included a table for comparative analysis, explicitly mentioning the methods adopted and the impact of those methods in terms of accuracy/improvements. Moreover, we have stated few working/deployed applications which reflects some shade of our idea of resume ranking.

## **Chapter 2 Project Vision**

In today's dynamic job market, the process of identifying the ideal candidate for a job position has become increasingly challenging and time-consuming. Recruiters are faced with the hectic task of going through a plethora of resumes, often resulting in inefficient and sometimes biased hiring decisions. Our project, "JobLens," envisions an innovative solution that uses modern technology to streamline and enhance the resume screening process, ultimately leading to better hiring outcomes.

### **2.1 Problem Domain Overview**

The problem domain in this project revolves around the challenges faced by recruiters and HR professionals in identifying the most suitable candidates for a job position. Nowadays, HRs have a lot of resumes available to go through, making it difficult to find the best candidate. Previous or already known ways to rank resumes are outdated as they only address specific areas of a resume moreover they only focus on semantics and how the resume is written, and to reach a definitive outcome the recruiters have to shortlist the resumes themselves.

### **2.2 Problem Statement**

Again the problem can be seen where there is a need to find and reach a more effective approach and solution to rank resumes based on the job descriptions. As already discussed above, going through a very large amount of resumes is not possible for a recruiter hence if an approach is devised which assists or automates the process to some extent will greatly help improve and speed up the hiring process. So the problem statement is that we are hoping to develop an intelligent resume ranking app which solves the above mentioned problem by ranking the resumes or at least recommending the resumes based on the criteria mentioned by the recruiters.

### **2.3 Problem Elaboration**

The problem can be elaborated by considering other problems that arise at the resume hiring process. With the repetitive task of going through 1000's of resumes recruiters also face the problem of general biasedness. Secondly, irrelevant resume applications take away a lot of time. With this the time to go through resumes of applicants that do not even fall under the job requirements can also be seen as a major problem for HRs. So as a whole if an application simply filters out the unwanted candidates and recommends or ranks the remaining candidates will make the hiring process more efficient and better

suited to the company's needs and this is what we aim to achieve.

## 2.4 Goals and Objectives

Ours goals and objectives are:

- Assisting with the initial process of going through the resumes to save time and effort for the recruiters
- Using Natural Language Processing to extract relevant sections such as skills, education, university etc. from a resume
- Aim to reduce human bias and errors in the process
- Develop a system that can accurately rank resumes based on the job description
- Provide options for recruiters to rank resumes based on a specific criteria

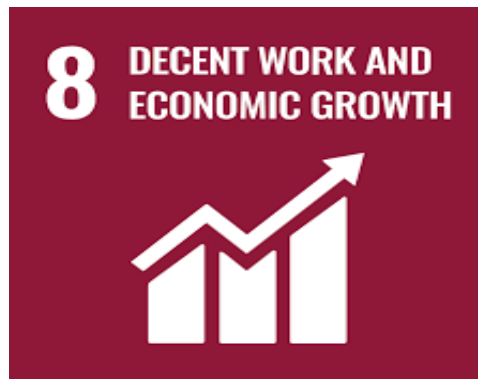
## 2.5 Project Scope

We are aiming to build a web application using React for the front end and Django framework for the back end. This application will use a large dataset of resumes to train the deep-learning model. The dataset will be collected from different sources, including job search websites, LinkedIn, social media platforms, and other online platforms. The dataset will be pre-processed and normalized to remove any unwanted information. Natural Language Processing (NLP) would then be used to extract relevant information from the data. Finally, a deep learning model will be developed and trained on the preprocessed dataset to identify the key features. The system will be tested using a sample dataset of resumes and evaluated based on its accuracy, efficiency, and usability.

## 2.6 Sustainable Development Goal (SDG)

### 2.6.1 Decent Work and Economic Growth

Our project covers decent work and economic growth sustainable development goals by helping organizations in their hiring processes. Improving the resume screening process and ranking the resumes afterward, can help businesses in saving time and resources in the HR department. This efficiency can lead to quicker job placements, which can therefore support economic growth by ensuring that individuals are employed in positions that match their qualifications.



**Figure 2.1: Sustainable Development Goal**

*This figure represents the SDG that is the target of JOBLENS*

## 2.7 Conclusion

In summary, JobLens represents a unique approach to address the challenges in the recruitment process, with the potential to make hiring decisions faster and beneficial for both organizations and job seekers in today's job market.

## Chapter 3 Literature Review / Related Work

The following section elaborates on the detailed literature review of previous research done in accordance with the chosen topic and gives detailed knowledge of each technique.

### 3.1 Definitions, Acronyms, and Abbreviations

The important definitions, acronyms and abbreviations used in this chapter are listed below:

- **BNB**: Boolean Naive Bayes
- **CV**: Curriculum Vitae
- **BiLSTM-CRF**: Bidirectional Long Short-Term Memory - Conditional Random Field
- **KNN**: K-Nearest Neighbors
- **SVM**: Support Vector Machine
- **MLP**: Multi-Layer Perceptron
- **LR**: Logistic Regression
- **NER**: Named Entity Recognition
- **TF-IDF**: Term Frequency-Inverse Document Frequency
- **SVG**: Scalable Vector Graphics
- **BERT**: Bidirectional Encoder Representations from Transformers
- **LDA**: Latent Dirichlet Allocation
- **PLSA**: Probabilistic Latent Semantic Analysis
- **CNN**: Convolutional Neural Network
- **POS**: Part-of-Speech
- **SER**: Speech Emotion Recognition
- **OCR**: Optical Character Recognition
- **BPNN**: Backpropagation Neural Network
- **CRNN**: Convolutional Recurrent Neural Network
- **IE**: Information Extraction
- **AIRANN**: Attentive Implicit Relationship-Aware Neural Network

## 3.2 Detailed Literature Review

This section covers notable literature published in the domains of Resume parsing using different techniques of Natural Language as well as ranking the resumes based on the given job description.

### 3.2.1 Resume Parsing Framework for E-recruitment

A resume parsing framework that enhances information accuracy by structuring unstructured resumes. It employs BNB and multinomial BNB for block classification and model training. Efficient resume screening is vital for effective recruitment strategies.

#### 3.2.1.1 Summary of the research item

The introduction of a resume parsing framework by Sajid et al [1], which takes an unstructured resume as an input and returns a data-enriched structured output to enhance the accuracy of extracted information, shifted the paradigm of resume parsing. After raw text extraction and block separation by text block recognition, data is enriched using ontology (a structured framework that defines relationships and categories within a specific domain). In the proposed pipeline, a probabilistic model named as Boolean Naive Bayes (BNB) algorithm is used to classify blocks of resumes [1], and a multinomial Boolean naive Bayes algorithm is used to train 5 language models that correspond to 5 classes i.e., education, skill, experience, interest, and personal information.

#### 3.2.1.2 Critical analysis of the research item

The strengths and weaknesses of this proposed approach are

- It improves the quality of information by enriching the data with ontology, which aids in defining connections and classifications within a certain domain.
- The framework's capacity to scale, which is crucial when dealing with a huge number of resumes, is not disclosed.

#### 3.2.1.3 Relationship to the proposed research work

This research paper is relevant to the first module of our project which includes the accurate parsing of resumes in order to get the exact information to increase the accuracy in ranking.

### **3.2.2 Design and development of a machine learning-based resume ranking system**

The KNN algorithm and NLP techniques are used to rate resumes according to how closely they match job descriptions. After data cleaning and TF-IDF vectorization, parsing efficiency was 85% and ranking efficiency was 92%.

#### **3.2.2.1 Summary of the research item**

In the Work of Tejaswini et al [2], NLP methods and the KNN algorithm are applied to rate resumes, assisting the firm in hiring the most qualified people. Firstly, candidates must pass a basic test then their resume will be ranked after data cleaning that primarily focuses on the removal of stop words and TF-IDF vectorization. K Nearest Neighbor machine learning algorithm will help in ranking the resume after calculating the cosine distance with the job description. 85% parsing and 92% ranking efficiency have been achieved [2].

#### **3.2.2.2 Critical analysis of the research item**

The strengths and weaknesses of this research paper are

- The approach employed is thoroughly described in the publication, along with data pretreatment procedures like stop word elimination and TF-IDF vectorization. Because of this openness, readers may reproduce the research and comprehend how the findings were reached.
- The paper does not address any potential drawbacks or difficulties related to the suggested strategy. To evaluate the method's suitability for use in real-world situations, it is important to be aware of its limits.

#### **3.2.2.3 Relationship to the proposed research work**

The paper relates more to the data-preparation module of our project which first applies data pre-processing techniques like stop words removal, POS tagging, etc and then followed by the parsing of resume

### **3.2.3 Resume analysis using machine learning and natural language processing**

By using ML and NLP approaches, the study seeks to improve recruiting. The algorithm achieves 95% parsing accuracy and 85% ML accuracy. It uses a variety of analyses and NER for data extraction, followed by an SVM and KNN-based Resume Ranking Algorithm.

### 3.2.3.1 Summary of the research item

In the Study by Jivtode et al [3], the main goal is to improve the recruitment process by leveraging machine learning and natural language processing techniques. The proposed method used Morphological, Syntactic, and Semantic analysis for the data extraction followed by Named Entity Recognition to get the most relevant data. After data extraction, the Resumes Ranking Algorithm comes under action, which mainly focuses on Support vector machines (SVM) and K Nearest Neighbors (KNN) to rank the parsed data using Machine learning algorithms. As a result, 95% accuracy in parsing and 85% machine learning accuracy is achieved[3].

### 3.2.3.2 Critical analysis of the research item

The strengths and weaknesses of the proposed approach are

- It Utilizes multiple analyses and NER for precise data extraction
- The paper doesn't discuss the practical aspects of deploying the proposed system in real-world recruitment processes,

### 3.2.3.3 Relationship to the proposed research work

The goal of our project and the research is to use machine learning and natural language processing (NLP) to enhance the recruiting process. They both want to discover the most qualified individuals by automating the evaluation of resumes.

## 3.2.4 Resumate: A Prototype to Enhance Recruitment Process with NLP-based Resume Parsing

The study focuses on teaching computers how to analyse textual texts with knowledge comparable to that of a human. It investigates several methods of named entity recognition, tokenization, text categorization, stemming, lemmatization, and uses a dictionary-based entity recognition system. SVM and XGBoost have among machine learning models the best predictive performance, as shown by cross-validation.

### 3.2.4.1 Summary of the research item

In the words of Mohanty et al [4], the aim is to train the computer to examine textual documents in a way that resembles a person. Numerous strategies, such as named entity identification, tokenization, text classification, Stemming, lemmatization, and approaches of other natural language processing techniques, have been investigated for evaluating and analyzing unstructured data [4]. A Dictionary-based

Entity recognition algorithm has been used. To assess a machine learning model's ability to predict fresh data, cross-validation is performed that reveals the high ranges of training performances of SVG and XGboost as compared to all other machine learning models.

#### **3.2.4.2 Critical analysis of the research item**

The strength and weaknesses of this approach are

- The study examines a variety of natural language processing methods that can help us fully comprehend unstructured textual material.
- The experiment's dataset size and other specifics are not specified, which might limit how broadly the results can be applied.

#### **3.2.4.3 Relationship to the proposed research work**

This approach focuses on applying natural language processing techniques to evaluate and analyze unstructured textual data and is relevant to our idea of parsing and ranking CVs.

### **3.2.5 Improved Resume Parsing based on Contextual Meaning Extraction using BERT**

This study shows that contextual comprehension using the BERT algorithm enhances resume identification above and beyond word matching. Following preprocessing and TF-IDF quantification, unsupervised machine learning using BERT is used to rank resumes with a 91% accuracy rate.

#### **3.2.5.1 Summary of the research item**

The existing methods will detect resumes based on word-matching approaches, but contextual understanding is crucial and is made feasible by the BERT algorithm, a study by Deepak et al [5]. After preprocessing and cleaning, including tokenization and parsing, the TF-IDF (Term Frequency- Inverse Document Frequency) metric helps to quantify the words. The main aim is to extract the data based upon contextual meaning that is possible by an unsupervised machine learning model called BERT, "Bidirectional Encoder Representations from Transformers." introduced at Google [5] that is trained on large amounts of textual data from Wikipedia and Book Corpus. The system employed by BERT for ranking resumes achieved 91% accuracy

#### **3.2.5.2 Critical analysis of the research item**

The strengths and weaknesses of this method are

- The technology goes beyond conventional word-matching techniques and makes use of the BERT contextual understanding algorithm, which can recognize subtleties and context in resumes, improving accuracy.
- The BERT model is computationally expensive to implement and train because it demands a lot of computing power and training data.

### **3.2.5.3 Relationship to the proposed research work**

Like our project idea, this approach emphasizes the importance of contextual understanding when parsing and ranking resumes. It goes beyond simple keyword matching and focuses on capturing the meaning and context of the content

### **3.2.6 Information Extraction from Text Intensive and Visually Rich Banking Documents**

A novel approach was introduced for extracting information from complex banking documents, combining deep learning techniques with both visual and textual data. They integrated this information extraction system into real banking environments and observed significant improvements, particularly with a 10% performance boost compared to prior methods.

#### **3.2.6.1 Summary of the research item**

Using both visual and textual data, the first work that employed deep learning to extract information from scanned unstructured banking documents that are both text and visually intensive was done by Oral et al [6]. The aim was to integrate the information extraction system for free formatted banking documents into a real banking environment. He observed the impact of using different neural word representations (BERT, Fast Text) and proposed a method that extracts named entities with a BiLSTM-CRF architecture and then uses a graph-based relation extraction algorithm to predict binary relations for each entity pair by a BiLSTM. He was able to achieve a 10-percentage point improvement as compared to previous approaches[6]

#### **3.2.6.2 Critical analysis of the research item**

The strengths and weaknesses of this research paper are

- The intention to incorporate the information extraction method into a genuine financial environment shows how practically useful the study is. It discusses a current issue and suggests possible remedies.

- Although visual data is mentioned in the study, no information about how it is handled and incorporated into the extraction process is given. The study would be more understandable if the visual data processing were covered in greater detail.

### **3.2.6.3 Relationship to the proposed research work**

This work relates to the information extraction module of our project that takes semi-structured data and outputs the structured version of it

## **3.2.7 Skill requirements in job advertisements: A comparison of skill-categorization methods based on wage regressions**

The study compares skill categorization methods using wage regression as an assessment measure, highlighting the effectiveness of Latent Dirichlet Allocation (LDA) in explaining 48.3% of wage variation from extensive job advertisement data.

### **3.2.7.1 Summary of the research item**

Ao et al [7] compare several skill categorization approaches using a new assessment measure based on wage regression, showing that the better skill categorization method should be able to extract the necessary skills from the job description more effectively with a dataset of 1158,926 job advertisements from the U.K. Performance of Word-counting methods with different dictionaries, as well as, different topic modeling techniques like LDA, PLSA, and BERTopic was compared. Experiments show that the best-performing method is the LDA with 48.3% of the wage variation explained [7].

### **3.2.7.2 Critical analysis of the research item**

The strengths and weaknesses of the proposed architecture in this paper are

- The study makes use of a sizable dataset of 1,158,926 job adverts from the United Kingdom, which improves the validity and applicability of the results. A benefit of using a large dataset is that it represents real-world circumstances.
- The chosen dataset, methodology, or evaluation measure may have some possible drawbacks or difficulties, but they are not covered in great detail in the work. By addressing these issues, the research might be evaluated more thoroughly.

### **3.2.7.3 Relationship to the proposed research work**

Our project involves the processing of resumes which primarily consist of the skills of candidates so the work of Ao can help in skills categorization

## **3.2.8 Intelligent Hiring with Resume Parser and Ranking Using Machine Learning and Natural Language Processing**

This study demonstrates an intelligent system that uses NLP and ML to rank resumes according to changing standards. It uses a thorough architecture that combines lexical, syntactic, and semantic analysis to improve recall and precision, creating a potent tool for matching resumes to job descriptions.

### **3.2.8.1 Summary of the research item**

An Intelligent system always ranks resumes of any format in accordance with the given constraints using NLP and ML as stated by Tiwari et al[8]. The main aim is to incorporate the changing parameters on which the resume will be ranked. The proposed architecture consists of a resume ranking system that performs Lexical analysis that segregates the resumes into blocks. The syntactic structure of the sentence is determined by syntactic analysis and after the generation of the parse tree with meaning of the sentence is determined by Semantic analysis. Precision, recall, and F-measure metrics are used for performance evaluation. The result is a supercharged tool for resume matching to job description [8]

### **3.2.8.2 Critical analysis of the research item**

The strengths and weaknesses of this paper are

- A noteworthy characteristic of the system is its capacity to adjust to shifting criteria for rating resumes. The value of certain abilities or credentials may fluctuate over time in a labour market that is dynamic, but our method seems to be able to handle these changes well.
- The scalability of the system is not discussed in the study. Effectively managing a big number of resumes and job descriptions is essential, especially in situations when there are a lot of job seekers.

### **3.2.8.3 Relationship to the proposed research work**

The study article and our project both center around the idea of ranking resumes. While the article proposes a method for categorizing and rating resumes based on multiple studies.

### **3.2.9 Semi-supervised deep learning-based named entity recognition model to parse the education section of resumes**

In this study, a semi-supervised technique is used to train a deep learning model for precisely identifying schooling portions in resumes. Named Entity Recognition (NER) tasks were completed with an excellent 92.06% accuracy. The procedure uses NER, CNN, and Bi-LSTM models, with an emphasis on prediction error reduction to improve educational information extraction.

#### **3.2.9.1 Summary of the research item**

Gaur et al [9] state that a semi-supervised approach is used to train a deep learning model to identify the education sections of resumes and then passed to the correction module to minimize the errors in predictions. The main objective is to accurately identify the educational information from resumes, that's why a dataset of 550 resumes is used after converting the pdf to JSON. Named entity recognition is used to identify all occurrences of a specific type of entity and then the adoption of CNN and Bi-LSTM served the task of prediction. Experiments show an accuracy of 92.06% on NER tasks [9].

#### **3.2.9.2 Critical analysis of the research item**

The strengths and weaknesses of this proposed methodology are

- It is advantageous to employ a semi-supervised technique since it enables the model to use both labeled and unlabeled data. This may result in improved generalization and more effective learning.
- An important strength is the presence of an error-correcting module. It solves the frequent problem of prediction errors, especially in NLP jobs. This module can improve the reliability and accuracy of the outcomes.
- There can be diversity restrictions depending on the source and kind of resumes in the collection. The data's representativeness could have an impact on the model's performance.

#### **3.2.9.3 Relationship to the proposed research work**

Both the research paper approach and our project relate in terms of data pre-treatment and extraction of data from the resume.

### **3.2.10 An automated resume screening system using natural language processing and similarity**

This method, which offers reliable vectorization and weighting algorithms, ranks resumes in accordance with how well they match job descriptions using semantic analysis and cosine similarity. It may overlook other crucial candidate aspects due to its primary ranking-focused approach, and for better performance, it could need a bigger dataset with a wider range of data.

#### **3.2.10.1 Summary of the research item**

Another approach used by Daryani et al [10] is to first extract the required information from an unstructured resume and then use cosine similarity to rank as per the job posting. The main goal is to develop a system that ranks the resumes in accordance with the distance to the job description. Tokenization, Stemming, Lemmatization, and POS tagging help in the preprocessing of unstructured text, and the result is summarized in JSON format. This system employs concepts of Vectorization, Importance/weight assigning techniques like TF-IDF, and lastly assigning rank by similarity calculation [10].

#### **3.2.10.2 Critical analysis of the research item**

The strengths and weaknesses of this research paper are

- The strategy places a strong emphasis on semantic analysis, which enables a deeper comprehension of the material in resumes. When comparing CV entries to job descriptions, this can produce more precise rankings.
- The model's capacity to generalize to a wider variety of resumes might be constrained by the usage of a dataset with just 550 resumes. For improved performance, larger and more diversified datasets are frequently required.

#### **3.2.10.3 Relationship to the proposed research work**

The first step in both our project concept and the research strategy is to analyze unstructured resumes to extract pertinent data. While the research focuses on collecting particular information for ranking, your proposal aims to gather a wider variety of data.

### **3.2.11 IntelliExtract: An End-to-End Framework for Chinese Resume Information Extraction from Document Images**

Research created "IntelliExtract," an end-to-end framework to enhance accuracy and speed in processing Chinese-language resumes.

### 3.2.11.1 Summary of the research item

Wang et al[11] created "IntelliExtract," an end-to-end framework to enhance accuracy and speed in processing Chinese-language resumes. The framework initially implements OCR that uses differential binarization and scene text recognition with a Single Visual Model to detect and recognize Chinese text with (SLANet and PicoDet) models used to identify tables in the resume. (VI-LayoutXLM) model is used to perform (SER). Finally (ERNIE) layout model is used to perform intelligent information filtering. Using 2000 manually created Chinese resumes from the Chinese Talent Resume Dataset, the collaborative model achieved over 95% cumulative accuracy. However, the author also states the chances of inaccurate results being produced when document image quality is low or there is complex formatting[11].

### 3.2.11.2 Critical analysis of the research item

The strengths and weaknesses of this paper are

- It integrates the use of OCR models, layout analysis, and information extraction to allow efficient processing of data.
- The accuracy of resume entity extraction has been improved by techniques like layout analysis and question-answer matching.
- System fails to accurately extract data from low-quality or complex format documents.

### 3.2.11.3 Relationship to the proposed research work

The document related to our work in terms of information extraction from the resume which is our module 1 of NLP. It employs different techniques to extract relevant information like name, place , skills, and address from a resume which we'll also do.

## 3.2.12 Automated extraction of information from Polish resume documents in the IT recruitment process

This research paper aims to utilize a multi-module system to increase the accuracy of information extraction from Polish linguistically oriented resumes.

### 3.2.12.1 Summary of the research item

Wosiak et al [12] aim to utilize a multi-module system to increase the accuracy of information extraction from Polish linguistically oriented resumes. The system processes unstructured CVs by segmenting

them into sections (e.g., education, work experience) and uses a hybrid module incorporating Linear2, Nerf, Bablefy, and CRF models. Additional services like Fuzzy Dictionary, AnchorNER for education data, and competence matching via Competence Lexem are employed, resulting in JSON-formatted output. Data for this research included binary PDF files provided by IT companies. The results show that the hybrid module performed 60 percent to 160% better than individual modules in identifying different keywords from different parts of resumes [12].

### **3.2.12.2 Critical analysis of the research item**

The strengths and weaknesses of the approach adopted in this paper are

- The approach initially divided the CV into major sections like education, and personal information and sent it to the appropriate model in their hybrid module.
- Identification of anomalies and handling outliers in various phases of information extraction is not included which might have produced better results

### **3.2.12.3 Relationship to the proposed research work**

The research paper uses NER tools which come under the Natural Language Processing (NLP) module of our project. We'll also be using NER tools to extract required data from resumes in order to prepare our dataset,

## **3.2.13 Human resources for Big Data professions: A systematic classification of job roles and required skill sets**

Paper studies Big Data's impact on skills via real job ads, identifying key roles and aiding hiring and planning.

### **3.2.13.1 Summary of the research item**

Mauro et al[13] studied Big Data's impact on skills via real job ads, identifying key roles and aiding hiring and planning. It initially starts with web scraping to obtain Big Data job posts. Then by text mining and expert judgment aid, it categorizes the job posts into 4 main categories. After that, it clusters similar skills and uses (LDA) Latent Dirichlet Allocation to model skills to a job description. Finally, it uses the LDA presence matrix to analyze a skill set probability to match with a certain job family. The dataset used was more than 2700 online job postings and the results conclude that Big Data jobs can be divided into 4 distinct families [13]

### 3.2.13.2 Critical analysis of the research item

The strengths and weaknesses of this approach are

- The paper utilizes Latent Dirichlet Allocation, LDA (Blei, 2012) with which each topic can be seen as a distribution over the dictionary of words included in the corpus of the documents under study. They did not just rely on a clustering model like K means because it does not cater to the complex set of competency requirements included in job posts.
- The dataset used is only US-based hence it is not sufficient to recognize global patterns in job postings.

### 3.2.13.3 Relationship to the proposed research work

The document utilizes an NLP technique Latent Dirichlet Allocation(LDA) which we can use in some phases of our project if it is required to summarize and categorize any textual data. However, this document is not really related to our idea or any of our major module(s).

### 3.2.14 Resume Parsing based on Multi-label Classification using Neural Network models

This paper makes a comparative analysis by judging the performance of varied neural networks in resume parsing.

#### 3.2.14.1 Summary of the research item

Liu et al [14] make a comparative analysis by judging the performance of varied neural networks in resume parsing. Models like CNN, BPNN, CRNN, and Bi-directional Long-Short term memory were trained to find the keywords in resumes and put labels on them. Jieba in Python was used to perform word segmentation. Resumes are analyzed by segmenting words and using a trained database to find contextually relevant high-scoring words. This process assigns suitable job positions through steps like segmentation, deactivation, and word substitution. CRNN excels with 96% parsing accuracy. BPNN is also effective (93%) but slow. The authors of this document suggest combining CNN with SVM (Support Vector Machine)/KNN for quicker domain-specific resume filtering [14]

#### 3.2.14.2 Critical analysis of the research item

This research paper evaluates itself under the following criteria:

- The system uses the power of various neural networks like CNN, BPNN, CRNN, and Bi-directional

Long-short-term memory in order to perform effective and more accurate resume parsing.

- The paper suggests adopting a hybrid model consisting of CNN combined Support Vector Machine, and CNN combined K Nearest Neighbor to minimize time and effort

### **3.2.14.3 Relationship to the proposed research work**

This document is very relevant to our area of work. We'll also be using neural network model(s) for both our NLP module and Resume Ranking module. Models like Bi-LSTM can be specifically used to perform NER(Named Entity recognition) task(s). Moreover, it also uses the Word2Vec technique to improve CRNN attention. Word2Vec might also be used in our project for resume parsing.

### **3.2.15 Resume Ranking based on Job Description using SpaCy NER model**

This document aims to make an NLP-based resume ranking system That Extracts the required information from the resume based on constraints and Ranks them. The system used (NER) to extract the required information and used the “Learning To Rank” technique to train the model and rank resumes relative to job criteria. The online tool “Dataturks” is used to train the custom spacy NER model. A total of 20 resumes were tested on the model and f-score and precision were calculated.

#### **3.2.15.1 Summary of the research item**

Satheesh et al [15] aim to make an NLP-based resume ranking system that Extracts the required information from the resume based on constraints and ranks them. The system used (NER) to extract the required information and used the “Learning to Rank” technique to train the model and rank resumes relative to job criteria. The online tool “Dataturks” is used to train the custom spacy NER model. A total of 20 resumes were tested on the model and f-score and precision were calculated [15]

#### **3.2.15.2 Critical analysis of the research item**

The strengths and weaknesses of this approach are

- The document termed the project as ‘Third Generation hiring system’ because it allows candidates to upload their resume in any format which is then converted to a predefined format.
- It uses a blank .spacy model which is trained on only 20 resumes. Hence the final accuracy of correct parsing and ranking is not satisfactory.

### **3.2.15.3 Relationship to the proposed research work**

The document is very close to our project idea. It uses NER for resume extraction which is our first module. Moreover, they prepared their small-scale dataset by manually annotating their CVs using an automation tool called ‘DataTurks’. We also aim to prepare our data set in the same way but at a larger scale.

### **3.2.16 DWIE: An entity-centric dataset for multi-task document-level information extraction**

The paper is focused on the DWIE dataset (Deutsche Welle corpus for Information Extraction). It employs Information Extraction subtasks to enhance the entity extraction and recognition process

#### **3.2.16.1 Summary of the research item**

Zaporojets et al [16] made a system that can make job acceptance/rejection decisions but has limitations. The paper introduces the DWIE dataset (Deutsche Welle corpus for Information Extraction), a novel multi-task dataset designed to enhance various aspects of Information Extraction (IE). This dataset combines four critical IE subtasks: Named Entity Recognition (NER), Coreference Resolution, Relation Extraction (RE), and Entity Linking. The paper suggests utilizing graph-based neural message-passing techniques between document-level mention spans [16]

#### **3.2.16.2 Critical analysis of the research item**

The strengths and weaknesses of the paper that we identified are:

- It combines multiple Information Extraction tasks, providing a diverse resource for researchers.
- The complexity in the information extraction pipeline may lead to practical challenges.

#### **3.2.16.3 Relationship to the proposed research work**

This research paper relates to the Natural Language Processing (NLP) module of our project specifically with its dataset preparation section. We will also be using Named Entity Recognition for preparing our dataset for the NLP module which is one of the techniques used in the paper.

### **3.2.17 Talent recommendation based on attentive deep neural network and implicit relationships of resumes**

In the paper, a two-stage resume recommendation model based on deep learning and attention mechanisms is proposed which is expected to improve and aid the recruitment process.

### 3.2.17.1 Summary of the research item

Huang et al [17] start by mentioning the hurdles faced by the companies while recruiting new talents. Due to the extensive amount of data, it is almost impossible to scrutinize and point out the perfect candidates. Therefore, they propose a solution to this problem. They suggest a dual-stage resume recommendation model which is based on deep learning and attention mechanisms which they referred to as the Attentive Implicit Relationship-Aware Neural Network (AIRANN) model. AIRANN model outperformed other baseline methods, showing a general improvement of 13.31 percent, 12.49 percent, 6.5%, and 7.17% over the state-of-the-art baseline under F1@6, F1@15, NDCG@6 and NDCG@15, respectively [17].

### 3.2.17.2 Critical analysis of the research item

After critically evaluating the paper we have come to the following conclusions:

- It highlights the importance of uncovering latent job preferences from hired employee resumes, an aspect often overlooked in previous research.
- While improvements in evaluation metrics are reported, their real-world results for recruitment outcomes are not provided.

### 3.2.17.3 Relationship to the proposed research work

The paper relates greatly to our resume ranking module. Firstly the goal of aiding and automating the recruitment process is in line with our goal. A machine learning model is implemented in the paper which is very similar to the design of our resume ranking model.

## 3.2.18 On-Demand Job-Based Recruitment For Organisations Using Artificial Intelligence

This paper highlights the importance of addressing employee attrition and proposes a solution to the problem using machine learning.

### 3.2.18.1 Summary of the research item

Jayakumar et al [18] aim was to devise an AI model to predict employee attrition and use those results to recruit suitable employees. Employee attrition refers to the loss of workers in a company. This study employs a data-driven approach using the IBM HR employee Attrition dataset. After cleaning the dataset, the Random Forest Algorithm predicts attrition. With this Python-NLP facilitates impartial

screening and sourcing. AI Recommendation aligns job needs with applicant profiles. This comprehensive methodology optimizes HR practices. After running the model on this dataset, an accuracy of 92.9% was achieved. The F1-score has been calculated to be 0.928, indicating a high level of performance [18]

### **3.2.18.2 Critical analysis of the research item**

After a comprehensive analysis of the paper we have reached the following outcomes:

- The paper addresses employee attrition, a highly relevant and significant issue, which can have a detrimental impact on organizations.
- The paper does not address potential biases in the data or algorithms and their impact on fairness in hiring.

### **3.2.18.3 Relationship to the proposed research work**

This paper has a very similar scope to our project, implementing natural language processing for screening and AI recommendation using machine learning. Both of these approaches are to be implemented in our project to achieve similar outcomes and efficient results.

## **3.2.19 Resume Parser and Summarizer**

This paper emphasizes the importance of resume parsing in streamlining the hiring process for HR departments and recruiters

### **3.2.19.1 Summary of the research item**

Khan et al[19] work on a resume parser. According to them, resume parsing streamlines the process of hiring and recruiting. It aims to convert the resume into structured data that helps to easily organize and search resumes according to the required skill sets. The dataset was gathered from websites such as Kaggle and GitHub. This dataset was used to train the AI model. The system finds various resume formats, extracts content from them, then parses and generates key-value pairs, it then sorts and stores them in a database, and produces a summary of the resume inputted[19]. Resume screening very challenging task for recruiters, as it involves analyzing different types of resume formats to find the right candidate.

### **3.2.19.2 Critical analysis of the research item**

The analysis of the research paper is:

- It highlights the possible future additions or improvements to the parsing process as a whole.
- The paper identifies a common problem recruiters face which is inaccurate resume parsing.

### **3.2.19.3 Relationship to the proposed research work**

This paper relates to our Natural Language Processing module, as the goal of the paper is to extract useful information from resumes to help the recruiters to hire the right candidates.

### **3.2.20 Resume screening using machine learning**

This paper addresses the significant challenge of resume screening faced by recruiters and presents an approach to make this process simpler

#### **3.2.20.1 Summary of the research item**

NAGA et al[20] try to streamline this process. A large dataset of resumes is used from Kaggle which is then preprocessed. After that Natural Language Processing (NLP) techniques like NLTK are used to extract key information from CVs. The system then employs various classifiers such as KNN, SVM, MLP, and LR for categorization. This approach results in a 97% accuracy in aiming to enhance accuracy and performance in the screening process [20].

#### **3.2.20.2 Critical analysis of the research item**

After reading the paper we have produced the following analysis:

- There's no mention of how the proposed system's performance was evaluated or validated, which is crucial for assessing its effectiveness.
- Employing various classifiers such as KNN, SVM, MLP, and LR increases the chances of finding the best-performing model.

#### **3.2.20.3 Relationship to the proposed research work**

The relation of this paper is to our Natural Language Processing module and they aim to screen and parse a diverse range of resumes, which falls under the scope of our NLP module.

### **3.2.21 Automated Tool for Resume Classification Using Semantic Analysis**

The research paper aims to develop a system in general to automate and simplify recruitment tasks for large industries.

#### **3.2.21.1 Summary of the research item**

Gopalakrishna and V. Varadharajanaim [21] aim to develop a system in general to automate and simplify recruitment tasks for large industries. Firstly, the required data is inputted by the HR team to the NLP

module. This module removes unnecessary information and outputs data in the form of tokens. Then the classification module analyzes the list of tokens in order to classify a resume into suitable domains. Finally, a graph is generated which points out the relevance of the candidate with respect to their domains. The efficiency of the voting-based classifier was 91.2% in predicting domains for the resumes in the test set[21]

### **3.2.21.2 Critical analysis of the research item**

This paper states the following pros and cons:

- The ensemble classifier performance is extremely good in comparison to individual classifiers with respect to prediction.
- The paper suggests that training deep learning models like Generative Adversarial Networks on large-scale data sets (structured and unstructured) can be more effective and reliable.

### **3.2.21.3 Relationship to the proposed research work**

The project is quite relevant to our work because it employs an NLP module to remove unwanted data and parse it into tokens. This is what our first module will be doing in its initial phase. Moreover, it also incorporated a classification part which is relevant to our second module on resume ranking.

### 3.3 Literature Review Summary Table

This table has a general overview of the papers that have been studied so far and their results.

**Table 3.1: Literature Review Summary Table**

*This table shows Title, publication year and different approaches proposed in the reviewed papers along results*

| Title                                                                                                                | Year | Method                                            | Results                                                 | D.No |
|----------------------------------------------------------------------------------------------------------------------|------|---------------------------------------------------|---------------------------------------------------------|------|
| Resume Parsing Framework for E-recruitment [1]                                                                       | 2022 | Text block recognition, Boolean Naive Bayes (BNB) | Accurate Extraction of information from resumes         | N/A  |
| Design and development of machine learning-based resume ranking system [2]                                           | 2022 | Cosine Similarity, KNN, TF-IDF vectorizer         | 85% parsing and 92% ranking accuracy.                   | 3    |
| Resume analysis using machine learning and natural language processing [3]                                           | 2023 | SVM, KNN                                          | 95% parsing and 85% ranking accuracy                    | N/A  |
| Resumate: A Prototype to Enhance Recruitment Process with NLP-based Resume Parsing [4]                               | 2023 | Dictionary-based Entity recognition algorithm     | High ranges of training performances of XGBoost and SVG | N/A  |
| Improved Resume Parsing based on Contextual Meaning Extraction using BERT [5]                                        | 2023 | BERT, TF-IDF vectorization                        | 91% accuracy in ranking                                 | 7    |
| Information Extraction from Text Intensive and Visually Rich Banking Documents [6]                                   | 2020 | OCR, BiLSTM-CRF                                   | 10% increase as compared to previous approaches         | 10   |
| Skill requirements in job advertisements: A comparison of skill-categorization methods based on wage regressions [7] | 2023 | LDA, PLSA & BERTopic                              | LDA outperforms PLSA and BERTopic                       | 9    |
| Intelligent Hiring with Resume Parser and Ranking Using Machine Learning and Natural Language Processing [8]         | 2016 | Lexical, syntactic and semantic analysis          | Tool for parsing and ranking                            | N/A  |

|                                                                                                                 |      |                                                                                                                                       |                                                                                                                                         |     |
|-----------------------------------------------------------------------------------------------------------------|------|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-----|
| Semi-supervised deep learning-based named entity recognition model to parse education section of resumes [9]    | 2020 | NER, CNN, Bi-LSTM                                                                                                                     | 92.06% accuracy in NER task.                                                                                                            | N/A |
| An automated resume screening system using NLP and similarity [10]                                              | 2020 | Vectorization, Cosine-Similarity, TF-IDF                                                                                              | Accurate ranking of resumes                                                                                                             | N/A |
| IntelliExtract: An End-to-End Framework for Chinese Resume Information Extraction from Document Images [11]     | 2023 | OCR, SLANet and PicoDet, VI-LayoutXLM, SER, ERNIE models                                                                              | Surpassed a threshold of 0.95 in terms of accuracy for extracting named entities from document images which are stated as “significant” | 4   |
| Automated extraction of information from Polish resume documents in the IT recruitment process [12]             | 2021 | Linear2, Nerf, Bablefy and CRF, Fuzzy Dictionary, Anchor NER competence lexeme service                                                | 60% to 160% better performance compared to separate common tools in identifying keywords from different parts of the resume             | N/A |
| Human resources for Big Data professions: A systematic classification of job roles and required skill sets [13] | 2018 | Web scraping, text mining, expert judgment, (LDA) Latent Dirichlet Allocation, LDA presence matrix                                    | Big Data jobs can be divided into 4 distinct families: Business Analyst, Data Scientists, Big Data Developers, Big Data Engineers       | N/A |
| Resume Parsing based on multi-label Classification using Neural Network models[14]                              | 2021 | CNN, BPNN, CRNN, and Bi-directional Long-Short term memory models for training, Jieba in Python was used to perform word segmentation | CRNN excels with 96% parsing accuracy. BPNN is also effective with 93% of parsing accuracy but is slow.                                 | 7   |

|                                                                                                         |      |                                                                          |                                                                                                                                                                                                           |     |
|---------------------------------------------------------------------------------------------------------|------|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| Resume Ranking based on Job Description using SpaCy NER model [15]                                      | 2020 | NER, Learning to Rank Technique, Dataturks                               | score of matching job description with skill set out of 3 java 1.0: 1/3 DE: 1/3 Java 2.0: 2/3                                                                                                             | N/A |
| DWIE: An entity-centric dataset for multi-task document-level information extraction [16]               | 2020 | Information Extraction Techniques Such as Named Entity Recognition (NER) | Improvements of up to 5.5 F1 percentage points                                                                                                                                                            | 8   |
| Talent recommendation based on attentive deep neural network and implicit relationships of resumes [17] | 2023 | Deep learning and attention mechanisms                                   | AIRANN model outperformed other baseline methods, showing a general improvement of 13.31%, 12.49%, 6.5%, and 7.17% over the state-of-the-art baseline under F1@6, F1@15, NDCG@6 and NDCG@15, respectively | 6   |
| On-Demand Job-Based Recruitment For Organisations Using Artificial Intelligence [18]                    | 2023 | Random Forest Algorithm, Python-NLP                                      | The F1-score of 0.928, indicating a high level of performance                                                                                                                                             | 2   |
| Resume Parser and Summarizer [19]                                                                       | 2023 | Python, Html, Css, Flask, NLP, Spacy                                     | N/A                                                                                                                                                                                                       | 3   |
| Resume screening using machine learning [20]                                                            | 2022 | NLTK, NLP, KNN, SVM, MLP, LR                                             | 97% accuracy in aiming to enhance accuracy and performance in the screening process.                                                                                                                      | 1   |
| Automated Tool for Resume Classification Using Semantic Analysis [21]                                   | 2019 | Ensemble learning, Dynamic Classification, Machine Learning, NLPP, KNN   | 91.2% accuracy in predicting domains for the resumes in the test set                                                                                                                                      | 5   |

### 3.4 Data Set Summary Table

This table has a general overview of the Available DataSet that have been explored so far.

**Table 3.2: Data Set Summary Table**

*This table shows the different available datasets and their statistics.*

| Sr.No | Name                                                                                 | Available At | Link                                                                                                                                                | Number of Resumes | Resume Types                                           |
|-------|--------------------------------------------------------------------------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|--------------------------------------------------------|
| 1     | Resume Dataset by Snehaan bhawal                                                     | Kaggle       | <a href="https://www.kaggle.com/datasets/snehaanbhawal/resume-dataset">https://www.kaggle.com/datasets/snehaanbhawal/resume-dataset</a>             | 2400+             | HR, Designer, Information-Technology, Teacher          |
| 2     | IBM HR Analytics Employee Attrition Performance                                      | Kaggle       | N/A                                                                                                                                                 | 1470              | Multiple                                               |
| 3     | Resume DataSet by Guarav Dutta                                                       | Kaggle       | <a href="https://www.kaggle.com/datasets/gauravduttakiit/resume-dataset">https://www.kaggle.com/datasets/gauravduttakiit/resume-dataset</a>         | 1700+             | Java Developer, Testing, DevOps, Python, Web Designing |
| 4     | Chinese Resume Dataset                                                               | Tianchi      | <a href="https://tianchi.aliyun.com/competition/entrance/231771/information">https://tianchi.aliyun.com/competition/entrance/231771/information</a> | 2000              | Multiple                                               |
| 5     | Dataset from Infosys                                                                 | N/A          | N/A                                                                                                                                                 | 30000             | Multiple                                               |
| 6     | Dataset from a company in Taiwan                                                     | N/A          | N/A                                                                                                                                                 | 5238              | Multiple                                               |
| 7     | Word2Vec                                                                             | Kaggle       | <a href="https://www.kaggle.com/datasets/sugataghosh/googleword2vec">https://www.kaggle.com/datasets/sugataghosh/googleword2vec</a>                 | N/A               | N/A                                                    |
| 8     | DWIE: An entity-centric dataset for multi-task document-level information extraction | github       | <a href="https://github.com/klimzaprojets/DWIE">https://github.com/klimzaprojets/DWIE</a>                                                           | 43,373            | Multiple                                               |
| 9     | U.K. job board dataset                                                               | Reed         | <a href="https://www.reed.co.uk/">https://www.reed.co.uk/</a>                                                                                       | 1,158,926         | Multiple                                               |
| 10    | Money transfer orders written in Turkish                                             | N/A          | N/A                                                                                                                                                 | N/A               | N/A                                                    |

## 3.5 Related Applications

This section covers the review of notable applications similar to the theme of our project.

### 3.5.1 Resume Worded

#### 3.5.1.1 Summary

Resume Worded [3.3] is an AI-powered platform that gives feedback on your resume and LinkedIn profile. It actually scores the resume based on the semantics or sentence structure. It gives you a line-by-line analysis and recommendations to improve the score. The most reassuring aspect of it is the "Magic Write" function, which substitutes AI-generated content for that which could result in low scoring. The dashboard maintains the logs of recent activities. Moreover, the premium version also helps you to identify the soft as well as hard skills.

#### 3.5.1.2 Critical Analysis

The strengths and weaknesses of Resume Worded are

- It provides in detailed analysis of a resume based on grammatical structure.
- It employs various grammar tests on distinct sections of resumes and displays the pass or fail status.
- By matching words rather than context, it analyzes the job description to find the abilities or keywords that are lacking from your resume.

#### 3.5.1.3 Relationship to Project

The main theme of Resume Worded relates to our project idea but to a small extent as it focuses on sentence mechanics instead of technical content. The dashboard provides the analytics which is similar to a major use case of our project.

### 3.5.2 LiveCareer

#### 3.5.2.1 Summary

LiveCareer [3.3] is a popular resume builder and career advice website. It offers a free resume review tool that scans your resume for over 30 issues, including grammar errors, typos, formatting errors, and keyword usage. The tool also provides personalized feedback on how to improve your resume and make it more competitive.

### **3.5.2.2 Critical Analysis**

The strengths and weaknesses of LiveCareer's resume review tool are:

- Scores your resume and highlights major problems found in it.
- It is very easy to use. The user just has to upload their resume.
- It is a free resource.
- Allows resumes to be uploaded in multiple formats.
- Allows you to improve your resume score by changing parts of your resume and giving suggestions.
- It is limited to checking for grammar and spelling errors, formatting issues, and missing keywords so no technical ranking is being performed.

### **3.5.2.3 Relationship to Project**

LiveCareer's resume review tool is similar to our project idea in a way that it provides feedback on resumes after parsing important information. However, our project focuses on the technical content of resumes for the purpose of ranking, while LiveCareer's tool focuses on grammar, typos, formatting, and keyword usage to rank/score resumes. Moreover, our project will allow multiple resumes to be uploaded whereas LiveCareer only allows one resume at a time since it does not perform any comparative analysis which we'll do in our project.

## **3.5.3 KickResume**

### **3.5.3.1 Summary**

The Kickresume Resume Checker [3.3] is an online tool that uses artificial intelligence (AI) to analyze resumes and identify areas for improvement. It can check for grammar and spelling errors, formatting issues, and missing keywords. It can also compare the resume to a database of successful resumes to identify areas where the user can strengthen the resume. In the end, it also scores the resume to give a comprehensive overview of how the uploaded resume ranks

### **3.5.3.2 Critical Analysis**

The strengths and weaknesses of LiveCareer's resume review tool are:

- Kickresumer is very easy to use. Simply upload your resume and the tool will do the rest.

- It provides comprehensive feedback on your resume, including specific suggestions for improvement.
- With this it also provides helpful tips on how to write a strong resume.
- It may not be able to identify all of the areas where your resume can be improved.
- Kickresume is limited to checking for grammar and spelling errors, formatting issues, and missing keywords. It does not provide feedback on the overall quality of the resume based on skills or other important areas

### 3.5.3.3 Relationship to Project

KickResume follows a theme similar to our project. It goes through the parsing phase to extract useful information and then identifies the areas where the user can strengthen the resume. Our project also envisions successfully parsing the uploaded resume and more importantly ranking it accurately at the end not only based on semantics and format but also on the provided information.

## 3.6 Applications Review Summary Table

This table has a general overview of the Applications similar to our project idea.

**Table 3.3: Applications Review Summary Table**

*This table shows the similar available applications and their limitations.*

| Sr.No | Name          | Link                                                                                                    | Limitations                      |
|-------|---------------|---------------------------------------------------------------------------------------------------------|----------------------------------|
| 1     | Resume worded | <a href="https://resumeworded.com/">https://resumeworded.com/</a>                                       | No contextual Analysis           |
| 2     | LiveCareer    | <a href="https://www.livecareer.com/resume/check">https://www.livecareer.com/resume/check</a>           | Superficial analysis of flaws    |
| 3     | KickResume    | <a href="https://www.kickresume.com/en/resume-checker">https://www.kickresume.com/en/resume-checker</a> | Analysis based on semantics only |

## 3.7 Conclusion

Natural language processing (NLP) and machine learning (ML) improvements have recently made advancements in resume parsing and ranking possible, and these developments have the potential to revolutionize HR and hiring procedures. These advancements make it possible to extract structured information from unstructured resumes and make it easier to rate candidates according to how well they match up with certain job requirements. Researchers have improved named entity identification, skill classification, and sentiment analysis, among other areas of resume analysis, by utilizing deep learning

models like BERT and CNN. NLP and ML approaches are adaptable to a range of recruiting difficulties thanks to their flexibility.

These innovations have an effect on recruiting and HR practices in the real world. Systems powered by AI are automating processes like resume screening, candidate rating, and even attrition prediction. This automation improves candidate selection accuracy while also saving time. Candidates are identified effectively thanks to models like the Attentive Implicit Relationship-Aware Neural Network (AIRANN). Additionally, BERT-based models have shown great potential in matching resumes with job descriptions utilizing semantic analysis and contextual comprehension. HR professionals stand to gain from enhanced efficiency and accuracy in applicant screening and staff management as these technologies continue to advance.

## **Chapter 4 Proposed Approach and Methodology**

In this chapter, we dive into the details of the methodology devised to solve the given problem. This chapter will primarily focus on the details of extensive dataset handling and basic pillars of the proposed approach. Starting with the collection of dataset from different source and pre-processing, this chapter will end at the heart of our study - the proposed methodology. We discuss the dataset items for both parsing and ranking for better understanding of the reader. As we navigate through various sections of this chapter, the understanding of the proposed approach and dataset prepared will be developed.

### **4.1 Dataset**

In this section, we will walk through the detailed exploration of the dataset for parsing and ranking module. Beginning with the overview of the data sources and walking through the description of each dataset item for both parsing and ranking. We elaborate over each dataset item to clearly define their state of usability within our research.

#### **4.1.1 Data Acquisition**

A variety of sources were consulted during the dataset's collecting process in order to provide an extensive and representative corpus for analysis. The Career Service Office (CSO), online repositories, and Linked-In—a professional networking site well-known for housing an abundance of professional profiles were the main sources of the dataset 4.1. All dataset items from these source sums up to 1000+.

##### **4.1.1.1 Internet Repositories**

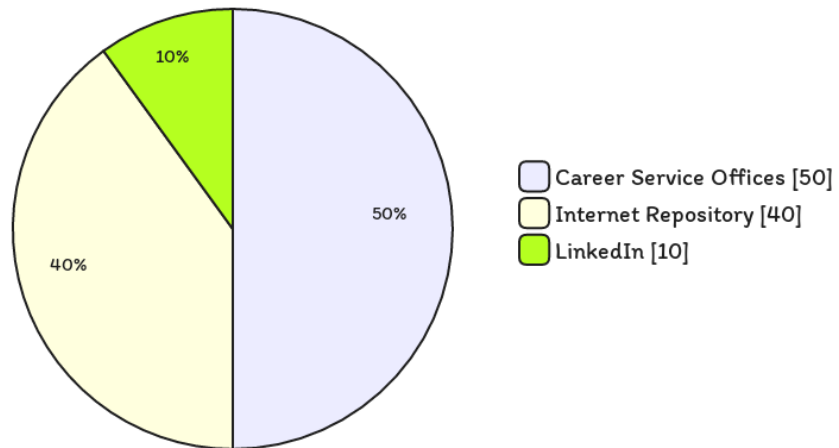
40% of the total dataset was contributed by the publicly available internet repositories. These repositories served as valuable resources of the dataset as they ensure the diversity in Resumes and reduce the need of data augmentation.

##### **4.1.1.2 Career Service Office (CSO)**

50% of total dataset was contributed by the Career Service Office (CSO) of National University of Fast and Emerging Sciences (NUCES). The dataset items were diverse in domains like there were resumes related to Computer Science Students, Software Engineering and even Electrical Engineering students.

### 4.1.1.3 Linked-In

The well-known professional networking site LinkedIn has become a valuable resource for finding modern resumes. Making use of the platform's accessibility, a focused strategy was used to collect profiles with a range of experiences and backgrounds in the workplace.



**Figure 4.1: Distribution of Data Sources.**

*This figure represents the distribution of data sources*

## 4.1.2 Dataset for Parsing Module

In this section details for dataset for parsing module has been presented. Manually labeling resume words is a gigantic task so a utility based tool was used to annotate the resume, one at a time. This tool helps to annotate or assign label to each word or a sequence of words from a resume. The NER (Named Entity Recognition) annotator can be found [here](#). This tool allows to make labels which can be defined once by the annotator. These labels can be used to categorize the information of the loaded resume data. Once the annotations are done, saving the file produces a JSON file which has the resume content, annotated words and their corresponding starting and ending index positions in the original resume text.

### 4.1.2.1 Dataset Item

Every dataset item is an object in which we have key words from the resume along with their tags, present at each index. The first prefix such as 'B', 'I', or 'U' represents the position or type of the tag. The second part represents the class of the tag such as skill, designation, degree, education etc. For example a label like 'B-INST' tell that the respective word token represents the beginning of the name of an word. The structure of an item is depicted in form of a table 4.1 for a better understanding.

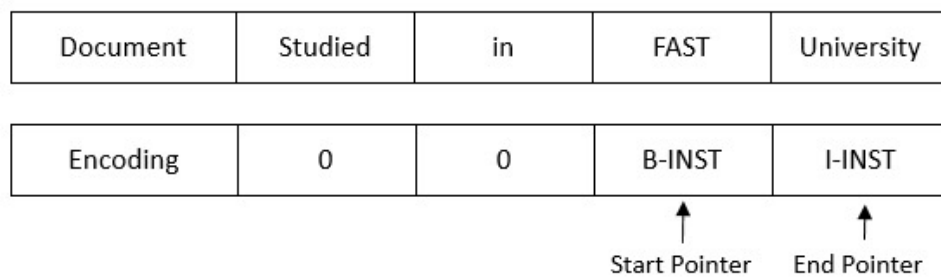
**Table 4.1: Parsing DataSet Item**

*This table represents a single dataset item for parsing module*

| Word       | Labels  |
|------------|---------|
| DevOps     | B-DESG  |
| Engineer   | I-DESG  |
| MySQL      | U-SKILL |
| CloudWatch | U-SKILL |
| Amazon     | B-SKILL |
| Elastic    | I-SKILL |
| CloudFront | U-SKILL |
| Ali        | 0       |
| BS         | B-DEG   |
| in         | I-DEG   |
| Computer   | I-DEG   |
| Science    | I-DEG   |
| Months     | 0       |

#### 4.1.2.2 BIO Encoding

Annotation of the dataset produced a JSON file containing word indexes and corresponding labels. A Python script was developed to extract words associated with each label index, followed by the implementation of the BIO encoding scheme 4.2. BIO encoding is advantageous for assigning a single label to word sequences, with "B" indicating the sequence beginning, "I" representing inside positions, and "O" marking positions outside the sequence. Its structured nature facilitates pattern generalization across sequences, enhancing the model's capacity to handle data variations and accurately label unseen or ambiguous sequences based on learned patterns.

**Figure 4.2: BIO Encoding Scheme**

*This figure represents the encoding of text belong to education section of a resume. Here B-INST and I-INST means Beginning and Inside of Institute respectively*

### 4.1.3 Dataset for Ranking Module

For this module the resume dataset was provided as an input by the parsing module. Moreover, multiple job descriptions of different domains from various platforms like Linked-In were gathered. Around 15 job descriptions were gathered for which 300 resumes were ranked manually by two annotators. This summed up to 4500 annotations. Moreover Cohen's Kappa score was calculated to find the level of agreement between two raters.

#### 4.1.3.1 Dataset Item

In this module, the dataset generated by the parsing module is used. Following preprocessing, a concatenated string of words for each original resume item generated by the parsing program was obtained. The similar procedure was used for job descriptions Table 4.2 shows resume text and job description for DevOps domain. Each of them is a combination of relevant words provided by the parser.

**Table 4.2: Ranking DataSet Item**

*This table represents a single dataset item for ranking module*

|                                                                                                                                                                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Job Description</b> : DevOps Engineer MySQL CloudWatch, Amazon Elastic CloudFront BS in Computer Science                                                                                                                                                                                  |
| <b>Resume</b> : aws services ec2 ecs lambda cloud formation cloudfront cognito s3 dynamodb rds route 53 amplify api gateways load balancers elastic search elastic cache jquery react linux docker devops engineer devops engineer web developer bachelor degree computer information system |

#### 4.1.3.2 Cohen's Kappa Score

Cohen's kappa coefficient, which ranges from -1 to 1, evaluates inter-rater agreement for categorical categories. Perfect agreement receives a score of 1, chance agreement receives a score of 0, while numbers below 0 indicate less agreement than chance. It assesses the degree of agreement among raters' categorization into categories that are mutually exclusive.

$$\kappa = \frac{P_o - P_e}{1 - P_e} \quad (4.1)$$

Where:

- $P_o$  is the observed proportion of agreement between raters.
- $P_e$  is the expected proportion of agreement between raters if agreement occurred by chance..

**Table 4.3: Cohen's Kappa Score**

*This table represents a Kappa Score for Different Job descriptions*

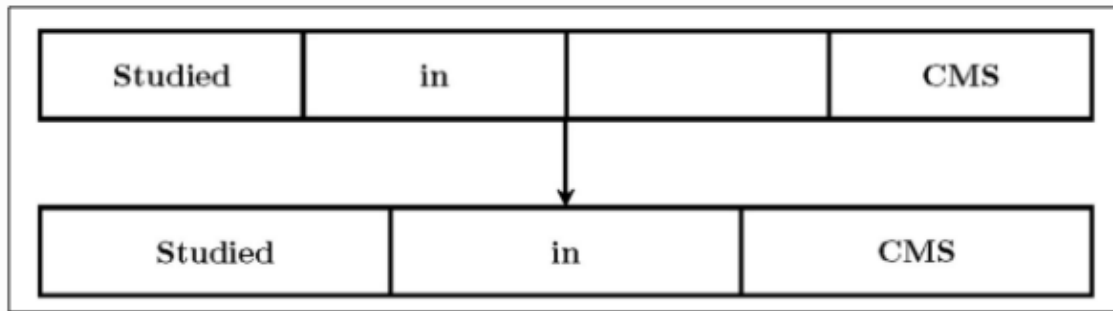
| <b>Job Description No</b> | <b>Kappa Score</b> |
|---------------------------|--------------------|
| 1                         | 91.7%              |
| 2                         | 91.20%             |
| 3                         | 88.89%             |
| 4                         | 95.62%             |
| 5                         | 95.62%             |
| 6                         | 87.55%             |
| 7                         | 85.86%             |
| 8                         | 91.43%             |
| 9                         | 82.73%             |
| 10                        | 88.89%             |

#### **4.1.4 Pre Processing of Dataset**

In this section we discuss pre-processing which is a critical step that is carried out to guarantee data uniformity, and readiness for efficient analysis before feeding the dataset into machine learning models. This multi step process involved cleansing, labeling, handling irregular spacing, correcting already labeled dataset items, tokenizing the text and padding the sequence etc. Moreover, removal of any non alphanumeric character and stop words was also done followed by lemmatization. Moreover, It also include BIO encoding technique because its structured nature facilitates pattern generalization across sequences, enhancing the model's capacity to handle data variations and accurately label unseen or ambiguous sequences based on learned patterns.

##### **4.1.4.1 Handling Irregular Spacing**

Words in a sequence with many spaces between them indicate irregularity, which could cause the model to analyze superfluous data while it is being trained. Optimizing model performance becomes dependent on how well this problem is handled. One way to deal with this problem is shown in the image 4.3. Through the use of a pre-processing step that combines multiple spaces into one, the model is freed from superfluous data to concentrate on pertinent information. By streamlining the input sequence, this method improves the model's efficacy and efficiency when handling textual data.

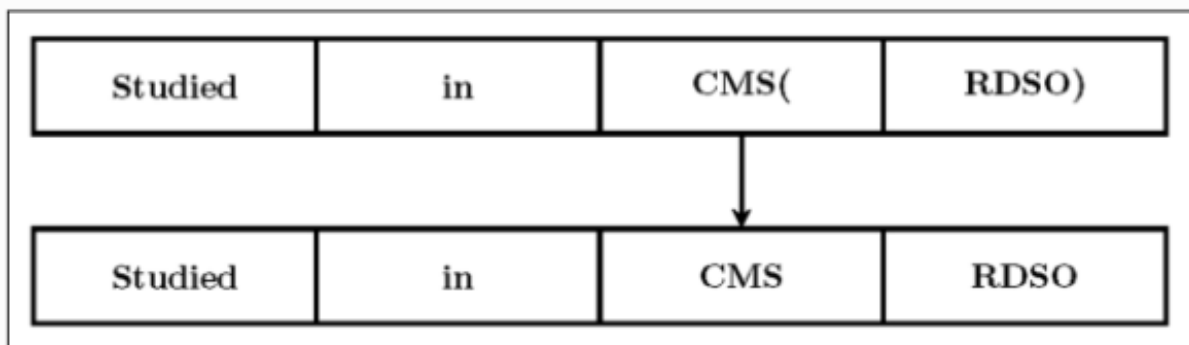


**Figure 4.3: Handling Irregular Spacing**

*This figure represents the removal of extra spaces between words*

#### 4.1.4.2 Resolving unbalanced Parenthesis

Unbalanced parentheses can cause gaps in datasets, especially when they are used to train machine learning models. These discrepancies may cause the model to interpret data in a biased way, resulting in less accurate or dependable results. Figure 4.4 depicts the strategy for dealing with this issue.

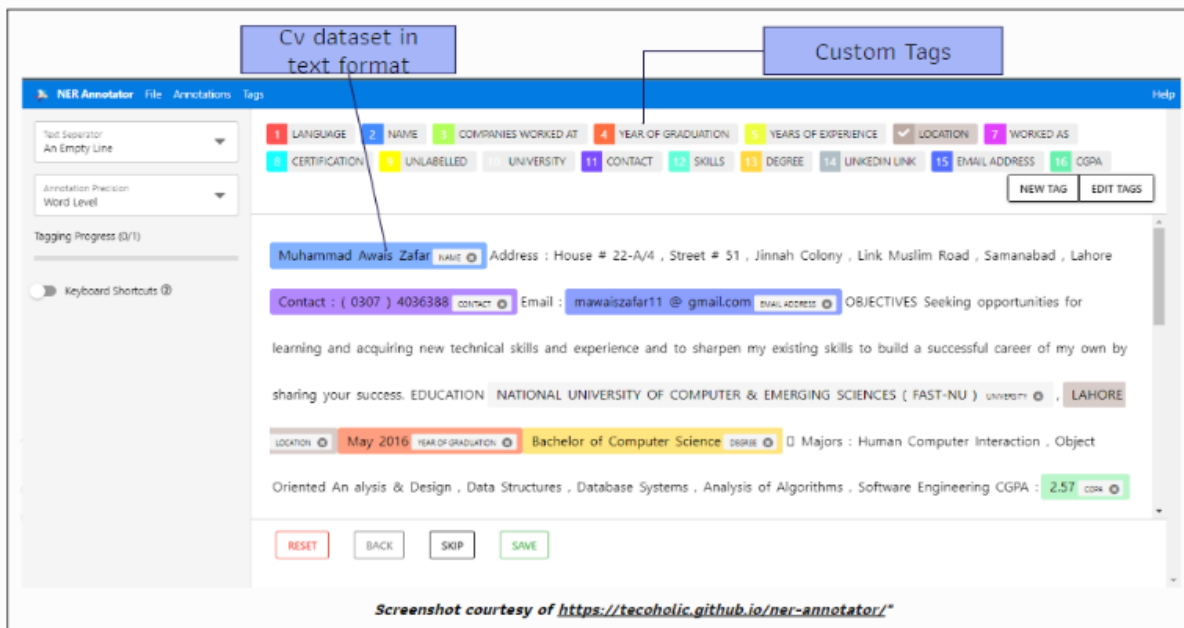


**Figure 4.4: Handling Unbalanced Parenthesis**

*This figure represents the removal of unbalanced parenthesis*

#### 4.1.4.3 Labeling of Dataset

For any supervised learning algorithm, labeling is one of the most important steps involved in the dataset preparation. Assigning labels to 1500 Resumes manually is itself a gigantic task, so a utility based tool was used to annotate the resume, one at a time. This tool helps us to annotate or assign label to each word or a sequence of words from a resume. The annotator can be found [here](#). Following picture explain the process of assigning labels to words from each resume



**Figure 4.5: Spacy NER Annotator**

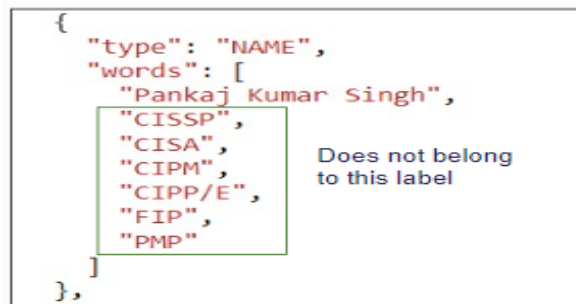
*This figure represents the process of labeling the words from a resume*

As an output, it will return a .JSON file that contains the indexes of the words and the labels for them. For further processing, a python script is written for two important tasks. First is to get the actual words from indexes against each label and second is to encode the labels using BIO encoding scheme, that is one of the classical encoding schemes in Natural Language processing. Beginning, Inside and Outside (BIO) encoding scheme is very useful when you want to assign a single label to a sequence of words. For the understanding, consider three pointers. First pointer as B (Beginning) will point to the starting word of the given sequence, second pointer as I (Inside) will point anywhere inside the sequence of words and last pointer as O (Outside) will point outside of the sequence to show the termination. Following picture explains this encoding technique for a sample text.

#### 4.1.4.4 Labeling Refinement

The dataset from the internet repositories was pre-annotated or pre-labeled but that contain many inconsistencies and wrong annotations. To use that dataset with the dataset from other sources, refinement in the labeling is necessary. This step will ensure the uniformity in the dataset by removing inconsistencies. Following are some examples of wrong labeling including labeling of Name, Awards, Degree, Companies Work At and Years of Experience etc. They are corrected by manually reading each resume and applying correction.

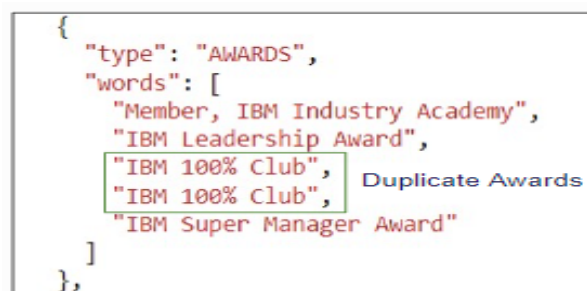
- 1. Name Label Refinement** Below picture shows the wrong labeling of text. The actual label is 'Name' but the words under this label are not expected.



**Figure 4.6: Labeling Refinement of Name**

*This figure represents the wrong labeling of text under 'Name' label*

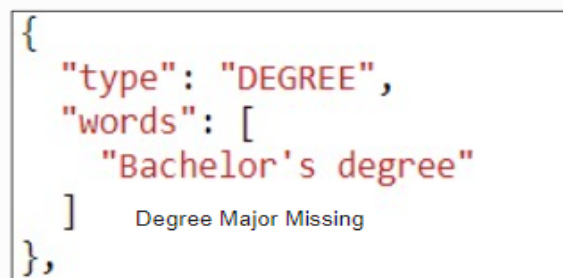
- 2. Awards Label Refinement** Below picture shows the duplication in the awards. Rewards should be unique in a dataset item.



**Figure 4.7: Labeling Refinement of Awards**

*This figure represents the duplicate awards.*

- 3. Degree Label Refinement** Below picture shows the missing of major in degree. This is corrected by manually reading the resume and adding text here



**Figure 4.8: Labeling Refinement of Degree**

*This figure represents the missing of major in degree.*

- 4. Companies Worked At Label Refinement** Below picture shows the wrong labeling of text under the tag of Companies Worked At.

```

{
  "type": "COMPANIES WORKED AT",
  "words": [
    "GrowByData",
    "1 year 6 months",
    "Omni Business Corporate International",
    "1 year 2 months",
    "Websurfer Nepal Communication System Pvt. Ltd.",
    "4 months"
  ]
},

```

Does not belong to this label

**Figure 4.9: Labeling Refinement of Companies Worked At**

*This figure represents the wrong labeling under tag of Companies Worked At.*

**5. Years of Experience Label Refinement** Below picture shows the wrong labeling of text under the tag of Years of Experience.

```

{
  "type": "YEARS OF EXPERIENCE",
  "words": ["3 years 10 months", "5 months", "2017"]
},

```

Does not belong to this label

**Figure 4.10: Labeling Refinement of Years of Experience**

*This figure represents the wrong labeling under tag of Years of Experience .*

## 4.2 Proposed Approach

In this section, a comprehensive overview of the proposed methodology for resume parsing and ranking is presented. The main goal is to create a sophisticated system that is capable to accurately extract the relevant information from the resume and subsequently rank them based on input criteria. The complex nature of resumes demands a sophisticated model architecture that is capable to capture the features as well as infer the relationship between different entities in order to assess the context in which entities are used.

### 4.2.1 Model Architecture for Parsing

The Deep learning model to solve the above stated problem consist of 5 layers including Word Embedding, Dropout, CNN, Bi-LSTM, and Dense Layer [22]. See Fig 4.11 for detail. Details of each layer is given below.

#### 4.2.1.1 Word Embedding Layer

The core element of our model design is the Word Embedding (WE) Layer. This layer converts textual data into dense vectors that represent the semantic connections between words. Each unique word in the vocabulary is assigned a unique vector in a high-dimensional space. For a resume, pre build word

embedding models cannot server the task as they lack vocabulary of this domain. So, we have created a new set of word embeddings based on our dataset in order to make the prediction accurate. Word Embedding Layers have been used for two different objectives as mentioned below.

### **1. Word Embedding Layer for Words**

Vocabulary of resumes may contain several words like degree programs, University Names, Projects etc. To extract all such words or entities from resume efficiently, understanding of the meanings of words is required which is a tedious task. WE Layer for Words makes this challenge easier by semantically analyzing the input and representing them as dense vectors of real numbers. The main advantage is that the words that are semantically related to each other will have dense vectors near to each other in dimensional space. For Example: Words like 'Bachelor's' and 'Degree' are semantically related to each other so their dense vectors will also be near in the n-dimensional space as compared to the words like 'gender' and 'Degree' that are not connected to each other meaningfully.

### **2. Word Embedding Layer for Parts-of-Speech (POS)**

Only the semantic analysis can not serve the purpose of accurate extraction of entities from resume because the grammatical structure of a sentence defines the basis of the semantics of words. To analyze the grammatical structure of a sentence, labelling of words with their corresponding parts of speech is required. This task is accomplished by the use of WE for POS. It first labels the words with parts of speech and then represent them as dense vectors based on assigned labels. For example: The dense vectors of the words that fall under the category of 'NOUN' i.e 'Degree' and 'Computer Science' will be much closer to each other in dimensional space as compared to the vectors of the words that belong to different categories.

So, the Combination of above mentioned WE layers for the semantic and syntactic analysis, forms the base layer of the model.

#### **4.2.1.2 Dropout Layer**

The Dropout Layer is essential for limiting overfitting and improving our model's capacity for generalization. This layer adds endurance by randomly discarding a portion of linkages during training. It allows model to increase the adaptability and performance on unseen data. The probability of drop out layer used is 0.3 as it experimentally give the best result.

### 4.2.1.3 CNN Layer

One of the main component of the proposed model architecture is the Convolution Neural Network Layer (CNN). The role of this layer is to extract and learn the local features of words. The used CNN layer is 1-D (Dimensional) as the input data is linear to extract further the character level features. CNN layer is essentially made up of filters or kernels that iterate over the input data and use convolution operators to get salient features. To introduce the non-linearity into the model, Rectified Linear Function (ReLU) is used. It works by transforming all negative input values to zero while leaving positive values unchanged hence introducing the simplicity in model and promoting faster convergence in training. The mathematical equation for the ReLU function is given below.

$$f(x) = \max(0, x) \quad (4.2)$$

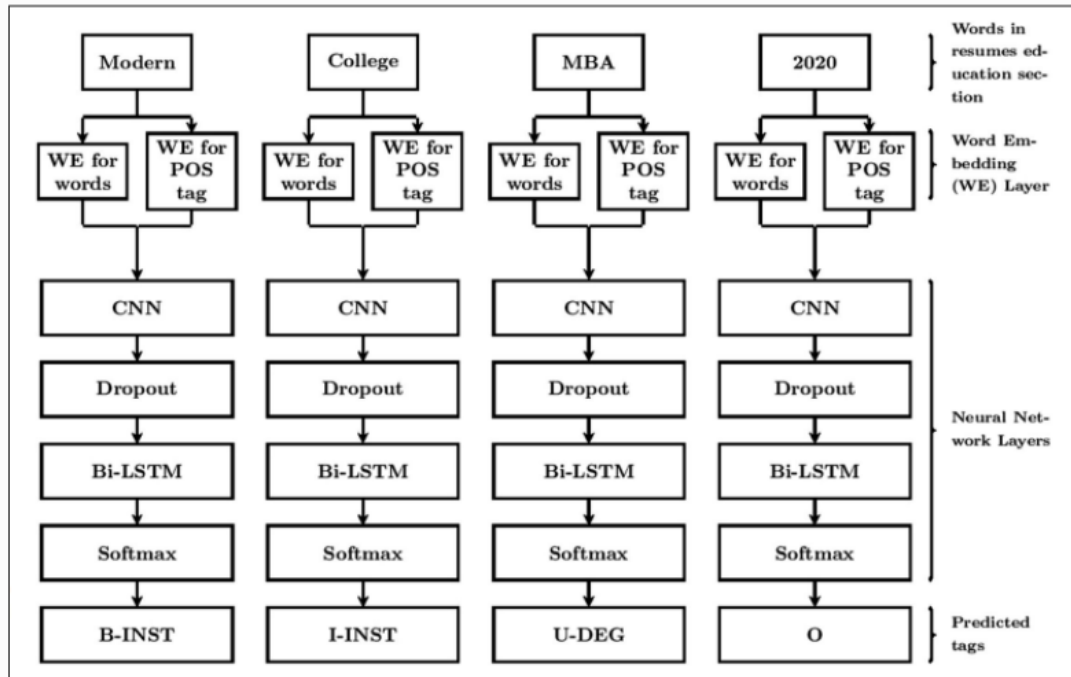
### 4.2.1.4 Bidirectional Long Short-Term Memory (Bi-LSTM) Layer

The Bidirectional Long Short-Term Memory layer is a Recurrent Neural Network (RNN) type layer that processes the input in both forward and backward direction. The mentioned unique feature allows this layer to play its vital role in the proposed model to grasp the context of the word in resume. In order to understand the text sequence, this layer captures the information from past and future states i.e for example the word 'Degree' will appear after the level of education i.e 'Bachelor or Master' and before the majors i.e 'Computer Science'. B-LSTM outperforms other traditional LSTM layers hence it is most suited option here. Further, this layer is concatenated with above layers to form base model.

### 4.2.1.5 Dense Layer

The dense layer - a fully connected layer is last component in the proposed model architecture. It connects each neuron in one layer to every neuron in the next layer, forming a densely connected neural network. This layer performs the linear operation followed by an activation function. The dense layer is constructed with the number of units equal to of unique classes and then concatenated with the previously formed model. The activation function used here is Softmax function that is widely in output layers of deep learning models. The mathematical Equation for the Softmax function is given below.

$$p_i = \frac{e^{z_i}}{\sum_{j=1}^n e^{z_j}} \quad (4.3)$$



**Figure 4.11: Model Architecture**

*This figure represents the architecture of the proposed model and can be found here [\[22\]](#)*

## 4.2.2 Architecture of Ranking Module

In this section we present our novel approach for ranking each resume based on a custom job description given by the recruiter. Here we employ our previously discussed NLP based parsing module to extract relevant information from resume. This information includes crucial information like skills, education, years of experience, GPA (if given), names of companies candidate has previously or currently working in, name of candidate, degree name and university name. Our technique leverages the Word2Vec model to represent both job descriptions and parsed resume text as high-dimensional vectors in a continuous vector space. Subsequently, cosine similarity is used to measure similarity between the two vectors.

### 4.2.2.1 Word2Vec Model

This model is used to represent words in a continuous vector space. It captures semantic relationships between words by learning vector representations that preserve the contextual information of words within corpus. Pre-trained Word2Vec model on a large corpus of text is used. This facilitated in capturing the nuanced meaning and context of words which are relevant to the respective domain and candidate. By transforming extracted information to numerical vectors enables quantitative analysis and comparison.

#### 4.2.2.2 Glove Model

The GloVe model (Global Vectors for Word Representation) is an unsupervised learning technique that generates word vectors based on co-occurrence statistics from a text corpus. GloVe use stochastic gradient descent or similar optimization methods to iteratively alter word vectors, which are subsequently represented as dense vectors in a continuous vector space called word embedding. GloVe can capture delicate semantic meanings. A co-occurrence matrix is built from the input text corpus to identify word relationships. The objective function is tuned to balance local and global co-occurrence statistics, resulting in convergence.

#### 4.2.2.3 Cosine Similarity

To quantify the similarity between the vector representations of the job description and the resume, cosine similarity is employed. Cosine similarity measures the cosine of the angle between two vectors which ranges from 0 to 90 degree. Greater the angle between the two vectors, the less similar they will be. For calculation of Normalized Cumulative Discounted Gain (NDCG) we have used this similarity for sorting of resumes after assigning scores. Formally, cosine similarity between vectors A and B is calculated as:

$$\text{cosine\_similarity}(\mathbf{A}, \mathbf{B}) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|} \quad (4.4)$$

### 4.3 Conclusion

In Conclusion, the suggested methodology offers a considerable solution to the challenges involved in extracting and ranking the resume information. The dataset is carefully handled, preprocessed, and cleaned using a multi-stage process to make sure it is well-structured and suitable for efficient model training. The use of sophisticated natural language processing methods, including tokenization and the BIO encoding scheme, makes it easier to express textual data meaningfully. The model architecture for parsing module, which includes layers for Word Embedding, CNN, Bi-LSTM, and Dense, was purposefully created to capture complex relationships and patterns in the resume data. Every layer makes a distinct contribution to the model's capacity to identify and categorize relevant information. Architecture of Ranking module circles around Word2vec model for creating embeddings of words and Cosine scores to measure similarity between input job description and resumes.

## Chapter 5 Implementation and Test Cases

In this chapter, details regarding Implementation of the project and various Test Cases have been provided. This chapter will primarily focus on details of implementation of proposed approach. Starting with the particulars of implementations, this chapter will end at the details of different test cases that fortify the suggested methodology.

### 5.1 Implementation

This section covers the implementation specifics of the project. It is further divided into following sub sections for each module of the project.

#### 5.1.1 Resume Parsing

The painstaking analysis of textual input via a well designed neural network architecture is the foundation of our resume parsing system. The procedure starts with the raw data preparation, which includes careful organizing and pre-processing to provide the best possible model training. The next important step in our process is to explore the world of word embeddings. Word embedding layers allow us to turn textual input into dense vectors that represent the semantic links between words. The resumes are converted into numerical representations in this initial step, which prepares the neural network's later layers. Next, we incorporate Convolutional Neural Network (CNN) layers into the architecture, which are positioned in a way that allows them to identify and retain the local structures of words in the resumes. This localized feature extraction improves the model's capacity to identify patterns and details in the text. Building on the local awareness that the CNN layers provide, the architecture smoothly moves to the domain of Bidirectional Long Short-Term Memory (BI-LSTM) layers, which provide a critical contextual understanding that allows the model to understand the relationships and sequential dependencies between words in resume. The bidirectional nature of the LSTM layers guarantees that the model takes into account both past and future context of each word, which further improves its capacity to perceive the grain level details. Lastly, we land at the Output layer i.e Dense layer with softmax activation function. The primary purpose of it is to convert the scores from Dense layers to probabilities.

#### 5.1.2 Resume Ranking

For the ranking module, we will accept a job description from the recruiter which will also be parsed by our parsing module to be matched with results of parsed resumes that include the skills, education, and degree sections. The matching algorithms utilized here will be to determine the similarity between resumes and job descriptions. Cosine similarity is used to measure the resemblance between two vectors

as it calculates the cosine of the angle between two non-zero vectors. In this case the higher the cosine similarity between the vectors of resume and the job description, higher the rank assigned to it.

We have employed mapping and mean vectors matching approach for resume ranking module after thorough testing and experimentation with different Word Embeddings models like Word2Vec, Glove and BERT. Among of all embeddings created, Glove gave most promising results. This implementation approach includes adding mapping for commonly used acronyms like "AI" or "JS" to 'Artificial Intelligence' and "JavaScript" respectively for both resume and job description. This normalization process has facilitated to overcome any inconsistencies of words between the resume and job descriptions and increase the quality of similarity scores between job description and resume text. This normalization increased the ranking efficiency and gave promising NDCG results.

## 5.2 Test Case Design and Description

As our project is research based so there it does not have traditional test cases. Here we provide sample test cases of each methodology with F1-Score as evaluation metric for given test cases.

### 5.2.1 Resume Parsing

This section provides a comprehensive description of the experiments conducted during the creation of the Resume Parsing Model. Considerable testing was conducted, utilizing a number of techniques. We have used F1 score as representative of our evaluation because it considers both the precision and recall providing a more balanced evaluation of our model's performance.

#### 5.2.1.1 Using WE + CNN

##### 1. Test Case No. 1

**Table 5.1: WE+CNN-Test case No.1**

*This is the table representing the actual text from a resume and BIO Encoded Predicted labels with F1-Score*

| Actual Text     | Predicted Labels | Actual Labels |
|-----------------|------------------|---------------|
| Python          | U-SKILL          | U-SKILL       |
| machine         | B-SKILL          | B-SKILL       |
| Learning        | O                | I-SKILL       |
| Computer        | I-DEG            | B-SKILL       |
| Vision          | O                | I-SKILL       |
| Air             | B-INST           | B-INST        |
| University      | I-INST           | I-INST        |
| <b>F1-Score</b> | <b>0.747</b>     |               |

## 2. Test Case No. 2

**Table 5.2: WE+CNN-Test case No.2**

*This is the table representing the actual text from a resume and BIO Encoded Predicted labels with F1-Score*

| Actual Text     | Predicted Labels | Actual Labels |
|-----------------|------------------|---------------|
| BS              | B-DEG            | B-DEG         |
| Computer        | I-DEG            | I-DEG         |
| Science         | I-DEG            | I-DEG         |
| C/C++           | U-SKILL          | U-SKILL       |
| Node.js         | U-SKILL          | U-SKILL       |
| Tailwind        | 0                | U-SKILL       |
| Git             | 0                | U-SKILL       |
| <b>F1-Score</b> | <b>0.911</b>     |               |

## 3. Test Case No. 3

**Table 5.3: WE+CNN-Test case No.3**

*This is the table representing the actual text from a resume and BIO Encoded Predicted labels with F1-Score*

| Actual Text     | Predicted Labels | Actual Labels |
|-----------------|------------------|---------------|
| Bachelor        | B-DEG            | B-DEG         |
| Of              | I-INST           | I-DEG         |
| Computer        | I-DEG            | I-DEG         |
| Science         | I-DEG            | I-DEG         |
| ec2             | U-SKILL          | U-SKILL       |
| S3              | U-SKILL          | U-SKILL       |
| AWS             | O                | U-SKILL       |
| National        | B-INST           | B-INST        |
| University      | B-INST           | I-INST        |
| of              | B-INST           | I-INST        |
| Computer        | I-DEG            | I-INST        |
| and             | B-INST           | I-INST        |
| Emerging        | B-INST           | I-INST        |
| Sciences        | B-INST           | I-INST        |
| <b>F1-Score</b> | <b>0.943</b>     |               |

### 5.2.1.2 Using BiLSTM+CNN with WE

#### 1. Test Case No. 1

**Table 5.4: BiLSTM+CNN+WE-Test case No.1**

*This is the table representing the actual text from a resume and BIO Encoded Predicted labels with F1-Score*

| Actual Text     | Predicted Labels | Actual Labels |
|-----------------|------------------|---------------|
| MERN            | B-SKILL          | B-SKILL       |
| Stack           | I-SKILL          | I-SKILL       |
| Kathford        | B-INST           | B-INST        |
| Engineering     | I-INST           | I-INST        |
| and             | I-INST           | I-INST        |
| Management      | I-INST           | I-INST        |
| College         | I-INST           | I-INST        |
| Bachelor's      | B-DEG            | B-DEG         |
| in              | I-DEG            | I-DEG         |
| Electronics     | I-DEG            | I-DEG         |
| and             | I-DEG            | I-DEG         |
| Communication   | I-DEG            | I-DEG         |
| Engineering     | I-DEG            | I-DEG         |
| <b>F1-Score</b> | <b>0.970</b>     |               |

#### 2. Test Case No. 2

**Table 5.5: BiLSTM+CNN+WE-Test case No.2**

*This is the table representing the actual text from a resume and BIO Encoded Predicted labels with F1-Score*

| Actual Text     | Predicted Labels | Actual Labels |
|-----------------|------------------|---------------|
| Python          | U-SKILL          | U-SKILL       |
| Machine         | B-SKILL          | B-SKILL       |
| Learning        | I-SKILL          | I-SKILL       |
| Computer        | B-SKILL          | B-SKILL       |
| Vision          | I-DEG            | I-SKILL       |
| Air             | B-INST           | B-INST        |
| University      | I-INST           | I-INST        |
| <b>F1-Score</b> | <b>0.952</b>     |               |

### 3. Test Case No. 3

**Table 5.6: BiLSTM+CNN+WE-Test case No.3**

*This is the table representing the actual text from a resume and BIO Encoded Predicted labels with F1-Score*

| Actual Text     | Predicted Labels | Actual Labels |
|-----------------|------------------|---------------|
| BS              | B-DEG            | B-DEG         |
| Computer        | I-DEG            | I-DEG         |
| Science         | I-DEG            | I-DEG         |
| Assembly        | U-SKILL          | B-SKILL       |
| Language        | I-SKILL          | I-SKILL       |
| SQL             | I-SKILL          | U-SKILL       |
| mongodb         | U-SKILL          | U-SKILL       |
| National        | B-INST           | B-INST        |
| University      | I-INST           | I-INST        |
| Of              | I-INST           | I-INST        |
| computer        | I-INST           | I-INST        |
| and             | I-INST           | I-INST        |
| emerging        | I-INST           | I-INST        |
| sciences        | I-INST           | I-INST        |
| <b>F1-Score</b> | <b>0.960</b>     |               |

## 5.2.2 Resume Ranking

In this section we discuss test cases that were performed to continuously evaluate and improve our ranking module. We used varying approaches as presented below. The one which gave the best results was selected to be used.

### 5.2.2.1 Using Baseline (BL) Approach

#### 1. Test Case No. 1

**Table 5.7: baseline-Test case No.1**

*This is the table representing the parsed resume text, parsed job description text and similarity angle*

|                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Job Description :</b> SQA Engineer testing strategie report defects Lead/collaborate Write code/scripts to test systems implementing test Computer Science Software Engineering SQL HTML ASPNET C# XML JMeter SDLC data driven testing grey box testing white box testing QMS ISMS TFS communication problem-solving |
| <b>Resume :</b> BS Software Engineering Black Box Testing SDLC C++ MySQL Data Structures Front End Testing Functional Testing Java OOP                                                                                                                                                                                  |
| <b>Angle :</b> 29.103                                                                                                                                                                                                                                                                                                   |

## 2. Test Case No. 2

**Table 5.8: baseline-Test case No.2**

*This is the table representing the parsed resume text, parsed job description text and similarity angle*

|                                                                                                                                                                                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Job Description</b> : Game Developer mobile games Unity Computer Science nreal C# C++ 2D and 3D game development game physics game mechanic game engines Object-Oriented programming Data Structures Algorithm multiplayer systems networking scripting Git AR/VR AI/ML Mobile game development |
| <b>Resume</b> : Bachelors of Computer Sciences [Computer Sciences] Internee ASPNET developer XHTML Java SQL ASPNet Unity Game Development                                                                                                                                                          |
| <b>Angle</b> : 42.516                                                                                                                                                                                                                                                                              |

## 3. Test Case No. 3

**Table 5.9: baseline-Test case No.3**

*This is the table representing the parsed resume text, parsed job description text and similarity angle*

|                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Job Description</b> : Devops Aws/gcp/huawei Cloud aws Huawei cloud Windows server Cloudflare Google Cloud Engineer huawei cloud and system Engineer Network Security Engineer Bachelor's degree Electronics and Communication Engineering                                                                                                                     |
| <b>Resume</b> :Cloud Engineer Gather client requirements cloud system architecture virtual servers cloud services O AWS Azure GCP Microsoft Exchange Servers Gmail/Google workspace Fileserver migrations SharePoint OneDrive optimize licensing documentation about cloud systems cloud infrastructure-related Timely upgrade Cloud Computing MCP MD MS AWS GCP |
| <b>Angle</b> : 36.90234375                                                                                                                                                                                                                                                                                                                                       |

### 5.2.2.2 Using Mean Vector Approach

## 1. Test Case No. 1

**Table 5.10: mean-vector-Test case No.1**

*This is the table representing the parsed resume text, parsed job description text and similarity angle*

|                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Job Description</b> : sqa engineer testing strategie report defects lead/collaborate write code-scripts test systems implementing test computer science software engineering sql html asp-net c# xml jmeter sdlc data driven testing grey box testing qc white box testing qms isms tfs communication problemsolving |
| <b>Resume</b> : bachelor software engineering black box testing sdlc c++ mysql data structures front end testing functional testing java oop                                                                                                                                                                            |
| <b>Angle</b> : 29.0863                                                                                                                                                                                                                                                                                                  |

## 2. Test Case No. 2

**Table 5.11: mean-vector-Test case No.2**

*This is the table representing the parsed resume text, parsed job description text and similarity angle*

|                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Job Description :</b> game developer mobile games unity computer science nreal c# c++ 2d 3d game development game physics game mechanic game engines object oriented programming data structures algorithm multiplayer systems networking scripting git ar/vr aiml mobile game development |
| <b>Resume :</b> bachelor computer science computer science internee aspnet developer xhtml java sql aspnet unity game development                                                                                                                                                             |
| <b>Angle :</b> 40.228                                                                                                                                                                                                                                                                         |

## 3. Test Case No. 3

**Table 5.12: mean-vector-Test case No.3**

*This is the table representing the parsed resume text, parsed job description text and similarity angle*

|                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Job Description :</b> cloud engineer gather client requirements cloud system architecture virtual servers cloud services aws azure gcp microsoft exchange servers gmail google workspace fileservers migrations sharepoint onedrive optimize licensing documentation cloud systems cloud infrastructure related timely upgrade cloud computing mcp md ms aws gcp |
| <b>Resume :</b> devops aws/gcp/huawei cloud aws huawei cloud windows server cloudflare google cloud engineer huawei cloud system engineer network security engineer bachelors degree electronics communication engineering                                                                                                                                          |
| <b>Angle :</b> 36.193                                                                                                                                                                                                                                                                                                                                               |

## 5.3 Conclusion

To Conclude, the project has two modules, Resume Parsing and Ranking Module. The resume parsing module, includes the Convolutional Neural Networks (CNN) followed by Bidirectional Long Short-Term Memory (BI-LSTM). These layers help extract the semantic and contextual meaning of the text which are finally integrated with the output layer i.e the dense layer with softmax activation, which converts the scores from the dense layer to probabilities. Our second module, i.e the ranking module requires two inputs, the already parsed resumes from parsing module and the job description given by the recruiter which will also be parsed in order to extract relevant words only. This module compares the similarity between these two text using cosine similarity and ranks the resumes based on the similarity with the job description. The higher the similarity of a resume with the job description, the higher it is ranked. Test cases mentioned in this chapter highlighted the actual and predicted labels from the resume in addition with the annotated labels assigned to the resume text.

## Chapter 6 Experimental Results and Discussion

In the following chapter we delve into the details of rigorous experimentation that was carried out to evaluate the performance of our modules at different stages. Moreover, we also present the respective results at different stages of development, which helped us to further optimize these results by re-evaluating our implementation techniques.

We've used multiple evaluation metrics to present the performance of our modules. We've mainly focus on F1 score for resume parsing results as it provides a better representation for this module by handling "O" class which causes unbalance. Moreover, we have employed NDCG(Normalized Discounted Cumulative Gain) to quantify the results for our ranking module.

### 6.1 Evaluation Metrics

In this section, various evaluation metrics are used to access the performance the proposed model. Details are given below.

#### 6.1.1 Accuracy

The accuracy of a classification model is one important measure of its general soundness. By dividing the entire amount of instances in the dataset by the number of cases that were correctly predicted, it is calculated. The formula for accuracy is expressed as

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} \quad (6.1)$$

#### 6.1.2 Precision

Precision measures the model's ability to forecast successful outcomes. It is calculated as the ratio of the number of true positive predictions to the sum of true positives and false positives. The precision formula is as follows:

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}} \quad (6.2)$$

#### 6.1.3 Recall

Recall, also known as sensitivity or true positive rate, evaluates the model's ability to identify all relevant occurrences. It is calculated as the ratio of real positives to the sum of false negatives and true positives. We may locate the recall formula by:

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \quad (6.3)$$

### 6.1.4 F1-Score

The F1-Score is the harmonic mean of recall and accuracy. Because it balances recall and accuracy, it is a helpful statistic in circumstances when the distribution of classes is not balanced. The formula for the F1-Score may be expressed as follows:

$$\text{F1-Score} = \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \quad (6.4)$$

### 6.1.5 Normalized Discounted Cumulative Gain (NDCG)

NDCG is a crucial indicator that provides deep insights into the efficacy of ranking algorithms. By taking into account both the order and the relevance of retrieved content, NDCG acts as a benchmark for evaluating the quality and relevance of search results. NDCG is calculated based on the relevance scores assigned to resumes in a ranked list. The formula for computing NDCG at a given rank  $k$  is as follows:

$$\text{NDCG@k} = \frac{\text{DCG@k}}{\text{IDCG@k}} \quad (6.5)$$

$$\text{DCG@k} = \text{rel}_1 + \sum_{i=2}^k \frac{\text{rel}_i}{\log_2(i+1)} \quad (6.6)$$

$$\text{IDCG@k} = \text{rel}_1 + \sum_{i=1}^{\min(k, |\text{relevant}|)} \frac{2^{\text{rel}_i} - 1}{\log_2(i+1)} \quad (6.7)$$

## 6.2 Experimental Setup

The flexibility and resilience of the resume parsing paradigm are strongly impacted by the experimental configuration. The model is trained across 15 epochs with a batch size of 10, and it uses the Adam optimizer to tune parameters so that it can adapt to changing data dynamics. In order to capture intricate patterns in the data and enable multiclass classification, activation functions such as softmax in the output layer and Rectified Linear Unit (ReLU) in the Convolutional Neural Network (CNN) layer were carefully selected.

For Resume Ranking we've experimented with various techniques to come to a conclusive point. We ended up using Glove model for the creation of embedding and cosine similarity to calculate the closeness between job description and resume text. However, several methods for the creation of word embedding like Word2Vec and calculating similarity like Euclidean distance were used to finally agree with the best option. Moreover, implementation techniques were also explored to improve our ranking results.

## 6.3 Results

The results were documented using an extensive suite of evaluation metrics, which allowed for a thorough investigation of the performance of each module. These results were essential for both reassessing the modules' efficacy and determine areas to be improved. The evaluation of the parsing model is based on four parameters: precision, recall, accuracy, and F1 score. However, we have chosen F1 score to scrutinize our parsing model results and identify any room for improvements. Normalized Discounted Cumulative Gain (NDCG) metric was used to evaluate the quality of the ranking algorithm. The formula for NDCG comprises of two parts: DCG (Discounted Cumulative Gain) and IDCG (Ideal Discounted Cumulative Gain).

### 6.3.1 Resume Parsing Results

This section contains the experimental results of the Resume Parsing model. Each model has been trained on the comprehensive dataset and classification report for each model are generated that helped to evaluate each model. The results are given in table 6.1

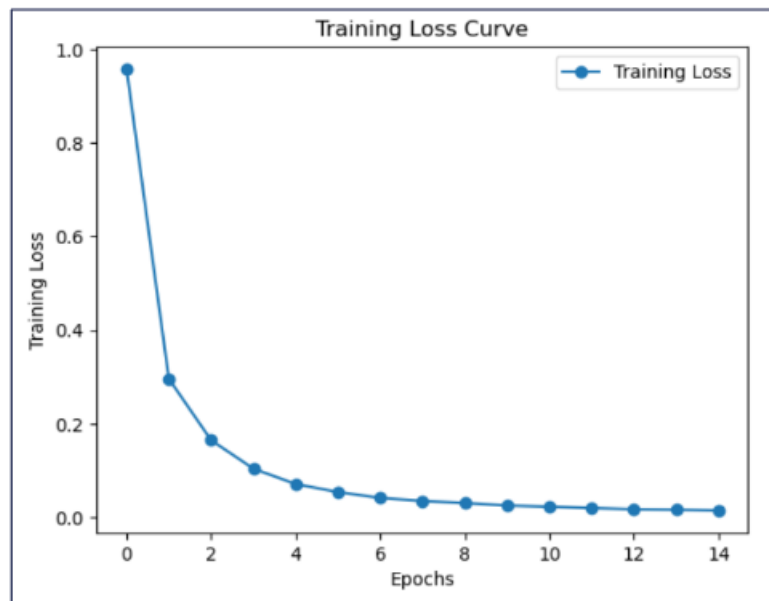
**Table 6.1: Resume Parsing Model Results**

*This table represents the results of different models including WE and Combination of WE with CNN and BiLSTM employed for resume parsing*

| Models            | Accuracy | Precision | Recall | F1-Score |
|-------------------|----------|-----------|--------|----------|
| WE + CNN          | 0.889    | 0.886     | 0.889  | 0.886    |
| WE + CNN + BiLSTM | 0.960    | 0.960     | 0.960  | 0.959    |

#### 6.3.1.1 Train Loss vs Epochs

One important part of the training process is visualizing the training loss to track the model's learning progress over epochs. The difference between the model's anticipated outputs and the actual target values during the training phase is known as the training loss or error. This measure is essential for determining how well the model fits the training set. Plotting the training loss over a series of epochs allows analysts to evaluate the performance of the model and spot patterns like divergence or convergence. Researchers can make well-informed decisions on model design, hyper-parameters, and optimization strategies to enhance overall performance by keeping an eye on the training loss. Furthermore, knowing the training loss helps identify problems like under- or over-fitting, which directs efforts to improve the model's ability to generalize to new data.

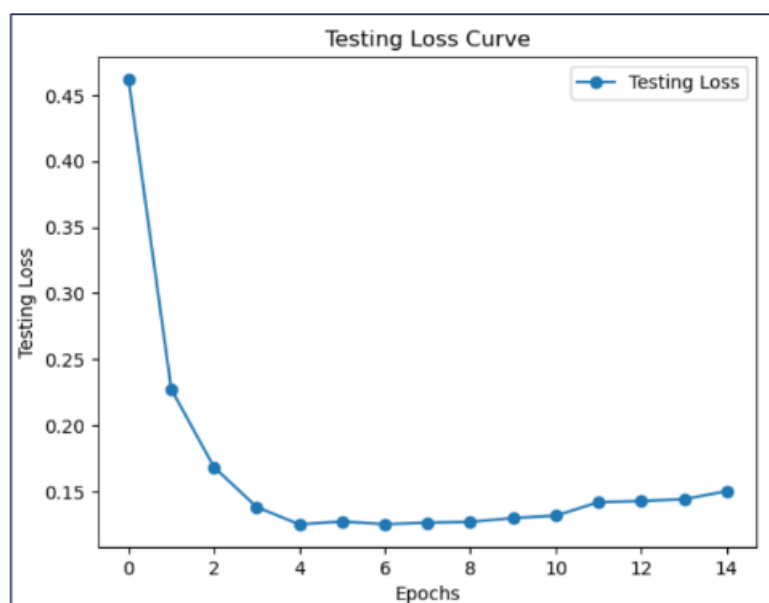


**Figure 6.1: Train Loss vs Epochs Graph**

*This figure represents a graph of loss in training vs epochs*

### 6.3.1.2 Test Loss vs Epochs

Analyzing the test loss against epochs graph is also essential for assessing how well the model performs with unknown data. A declining test loss across epochs indicates that the model is learning to accurately predict new, unseen instances instead of memorizing the training data. Test Loss Graph vs Epochs is given below



**Figure 6.2: Test Loss vs Epochs Graph**

*This figure represents a graph of loss in testing vs epochs*

### 6.3.2 Resume Ranking Results

This section contains the experimental results of Resume Ranking Model. To develop an efficient Resume Ranking Model, we tested various techniques, focusing on accuracy and relevance. We used Cosine Similarity with Glove model to capture semantic relationships and textual complexity.

#### 6.3.2.1 Baseline (BL) Approach Results

First of all removal of stop words was incorporated in pre-processing. Moreover the tags provided by the resume model were also used to only utilize the necessary data. In this case only the labels of skills, education and degree were extracted from both the resume and job description to remove unnecessary words resulting in a more accurate and cleaner version of the texts. The NDCG score obtained after evaluating the results using this approach was around to 0.7. The NDCG score can be assigned between the range of 0 to 1 so in this case the was closer to 1 which highlighted the relevance of the assigned ranking to the true ranking in the dataset. The table 6.2 depicts the results.

**Table 6.2: BL-NDCG Score Table**

*This table represents NDCG score of 10 Job Descriptions at different cut of points under baseline approach.*

| Job Description No | NDCG@5       | NDCG@10      | NDCG@15       | NDCG@20       |
|--------------------|--------------|--------------|---------------|---------------|
| 1                  | 0.879        | 0.877        | 0.877         | 0.938         |
| 2                  | 0.458        | 0.501        | 0.48          | 0.513         |
| 3                  | 0.138        | 0.234        | 0.32          | 0.32          |
| 4                  | 0.75         | 0.75         | 0.75          | 0.75          |
| 5                  | 0.438        | 0.424        | 0.553         | 0.553         |
| 6                  | 0.856        | 0.755        | 0.801         | 0.801         |
| 7                  | 0            | 0.147        | 0.147         | 0.147         |
| 8                  | 0.457        | 0.457        | 0.457         | 0.457         |
| 9                  | 0.796        | 0.947        | 0.947         | 0.947         |
| 10                 | 0.5          | 0.5          | 0.5           | 0.5           |
| <b>Average</b>     | <b>0.572</b> | <b>0.559</b> | <b>0.5832</b> | <b>0.5926</b> |

### 6.3.2.2 Mean Vector (MV) Approach Results

A final attempt was made to further improve the results. To increase the similarity between the job description and resume text normalization was integrated in this approach. Due to further improvement of the input text the distance between the vectors decreased and hence the NDCG score achieved was around 0.85. The results of this approach are shown in the table 6.3.

**Table 6.3: MV-NDCG Score Table**

*This table represents NDCG scores of 10 Job Descriptions at different cut of points under mean vector approach.*

| Job Description No | NDCG@5        | NDCG@10       | NDCG@15       | NDCG@20       |
|--------------------|---------------|---------------|---------------|---------------|
| 1                  | 1             | 0.845         | 0.9           | 0.954         |
| 2                  | 0.823         | 0.88          | 0.78          | 0.702         |
| 3                  | 0.598         | 0.598         | 0.598         | 0.733         |
| 4                  | 0.95          | 0.95          | 0.95          | 0.95          |
| 5                  | 0.739         | 0.666         | 0.585         | 0.672         |
| 6                  | 1             | 0.949         | 0.986         | 0.986         |
| 7                  | 0.86          | 0.643         | 0.745         | 0.789         |
| 8                  | 0.532         | 0.633         | 0.603         | 0.668         |
| 9                  | 0.89          | 0.966         | 0.966         | 0.966         |
| 10                 | 0.421         | 0.421         | 0.421         | 0.421         |
| <b>Average</b>     | <b>0.7831</b> | <b>0.7551</b> | <b>0.7534</b> | <b>0.7841</b> |

## 6.4 Conclusion

To conclude, we conducted plenty of experiments for our modules in order to develop a defined methodology. For resume parsing we tested its robustness and performances based on certain metric. Moreover, we considered different approaches in defining its model in terms of its layers and data set it is trained on to get the best results. For ranking module, its performances is mainly based on the ability to find in depth similarity between the two texts. All of these results have been portrayed with the help of statistical figures for a deep insight of experimentation and results section.

## Chapter 7 Conclusion and Work Completed

This chapter is divided into two sections: Conclusion and Work Completed. Finally, this chapter summarizes the research findings. It also provides a detailed analysis of the results obtained. The section on completed work discusses the preparation of dataset both for parsing and ranking module.

### 7.1 Conclusion

To sum up, the implementation of the parser module has shown that it is a reliable and efficient method for extracting information from resumes. By utilizing CNN layers, Bidirectional LSTM layers, and word embeddings, the model demonstrates a remarkable capacity to identify local and contextual structures present in the textual input. When a dense layer is used for multi-class classification, resume text may be accurately parsed into predefined categories. The assessment section discusses how the parser module obtains competitive performance metrics through thorough training on a variety of datasets. After rigorous testing and evaluating the model multiple times an impressive accuracy of 96% was achieved. As the foundation for the ranking component that comes after it, parsing module makes sure that relevant information is adequately captured from resumes.

The approach continues beyond the parsing step to rank and prioritize processed data, which brings us to our emphasis on the ranking module. The model makes use of sophisticated techniques, including deep learning, to comprehend the complex semantic and contextual elements included in resumes. When neural network architectures are combined, a single system is created that is capable of parsing, ranking, and overall resume analysis and evaluation. The report's conclusions show how effectively the ranking module provides relevant and tailored job recommendations based on resume content. Moreover multiple approaches were applied in this model and it was tested and evaluated accordingly. As evidenced by our analysis using the Normalized Discounted Cumulative Gain (NDCG) metric, an average NDCG value 0.85 was achieved which signified the relevance of the model's assigned rankings to the original ranking.

### 7.2 Work Completed

First of all a dataset was prepared by annotating 1000+ resumes manually, in which fields such as university, education, degree, name, skills were annotated. Moreover a ranking dataset of around 15 job descriptions was prepared and Cohen's Kappa Score was computed to rate evaluations of around 4500 manually assigned rankings. The parsing module was trained to extract fields such as phone, name, linked in, email address, degree and institute name. It was experimented using the word embedding

model and an accuracy of 89% was achieved. To improve the model CNN and Bi-LSTM layers were added to the previous model accuracy of 96% was achieved. Finally a ranking module was developed to calculate similarity between the job description text and resume text. It was experimented on various techniques, such as cosine similarity, euclidean distance, Manhattan distance etc. Cosine similarity gave the most promising results out of all the techniques. Moreover the Word2Vec and Glove models were integrated in the module to produce vectors for resumes and job descriptions. However, the most promising results were given by Glove model hence it's incorporation led to an average NDCG score of 0.85.

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